

edet

A non-hoardable clearing medium in which credit capacity
is a minimum cut over declared underwriting

the *edet* project

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Abstract

edet is a clearing medium for communities of mutual producers. Debt originates peer-to-peer at the point of sale, discharges by circulating onward as its bearer delivers value, and cannot be held: no token, no bearer instrument, no hoardable store of value. A member accumulates dated claims on named debtors, and *capacity* — bounded, non-transferable, and not a possession, since it lives in the stakes others have placed on them.

Capacity is the **maximum flow into an account from the community's underwriters**, across directed stakes recording what creditors have placed on one another. Nobody can stake beyond what they may confer, and outstanding credit reserves the flow that justified it. The ledger keeps stakes and obligations and computes the cut, so there is no score to award, adjust or revoke. An underwriter is any member who has declared a supply, an accepted liability seated only by a ceremony: genesis, or an amendment the community endorsed. The role is open, so a community is not limited to its founders; nor is it enriched from inside, since a supply declared against capacity earned within the community would reallocate credit without creating any — what a community can carry is its seed plus what later underwriters bring from outside. Admission, reputation and credit limits are that one number, with no attestation, probationary tier, sponsorship, admission or creation transition: an account is seated by the first bonded trade that names its key.

Sybil resistance follows from max-flow/min-cut rather than from a rule: the capacity of any *set* of accounts is bounded by the edges crossing into it. Minting identities confers nothing, a colluding member passes on no more than they hold, fabricated volume never crosses the boundary of the set that fabricated it and so is arithmetically zero with nothing to detect, and attack costs exactly what participation costs.

Two structural commitments carry the rest. *Capacity underwrites, it does not permit*: credit beyond a debtor's capacity is allowed but uninsured, borne by the creditor alone, which lets a community start from zero without a starter allowance for anyone who can generate keys. *Ledger-scoped total order*: a ledger is one replicated state machine under Byzantine-fault-tolerant consensus among validators its own members operate, with no world-global chain, no gas and no fee token. A ledger is not a community: a community is a region of the stake graph, the boundary between two is the absence of a path, one ledger carries as many as its members form, and there is no bridge.

The guarantees are invariants audited over the whole state after every block: the cut bound on every account set, reservations conserved against outstanding credit, no stake above what its creditor could confer, neither decay nor withdrawal able to age collateral out from under a live debt, and loss mutualisation bounded by what each underwriter declared.

Two boundaries remain, both economic, and acceptance is not one of them: a sale discharges before it creates, so a member who owes is paid by ceasing to owe. What binds is what makes clearing possible at all, a sufficient share of what a community buys being available from within it — a property of the community, ascertainable before it founds anything. And zero is absorbing: a community that underwrites nothing at genesis can trade on uninsured credit indefinitely, but can leave that state only through a founding ceremony.

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1 Introduction and design position

1.1 What *edet* is

edet is a clearing medium for a community of mutual producers: people and small enterprises who buy from one another regularly enough to clear their obligations against each other instead of settling in national currency.

Its unit is a debt contract, created by a sale from buyer to seller in a community unit. It discharges when the buyer delivers value onward — to the seller, or to anyone the seller directs it to — so an obligation moves through the community as production does and is worked off rather than paid out of a stock. There is no token, no bearer instrument, no balance and nothing to hoard: a large claim is a large expectation of future delivery, not a holding. What a member accumulates is *capacity*, a ceiling on how much the community will underwrite them for: bounded, non-transferable, not for sale, living in the stakes others have placed on them and falling when they withdraw (§4).

The design is measured for communities up to the order of 10^5 accounts on one ledger (§12); a larger ledger is an open problem this paper states rather than escapes, since a stake graph cannot be split without severing every edge across the cut.

1.2 The two structural commitments

Capacity is a minimum cut. A member’s capacity is the maximum flow from the community’s underwriters to their account, across directed stakes recording what creditors have placed. Nobody can stake beyond what they may confer, outstanding credit reserves the flow that justified it, and an underwriter is any member who has declared a supply they will stand behind. The ledger stores stakes and outstanding obligations; capacity is the cut computed from them.

One equation settles four questions a credit system normally answers with policy. Admission: an account with no stakes has capacity zero, so creating one grants nothing. Reputation: the stakes *are* the reputation. Sybil resistance: the capacity of any set of accounts is bounded by the stakes crossing into it, so identities are free and worthless together. Fraud: volume internal to a set never crosses that set’s own boundary, so fabricated trade is arithmetically void. These are theorems rather than process — Theorem 8.2 is argued once — where a rule that must be remembered can be forgotten, one that must be enforced can be evaded, and a bound re-argued for each new attack is eventually re-argued wrongly.

Capacity underwrites, it does not permit. Credit within a debtor’s capacity is *insured*: it reserves flow, and the community’s recourse machinery stands behind it. Credit beyond it is permitted but *uninsured*, and the creditor bears it alone. This is what lets a community begin: first trades are uninsured, they settle, they create stakes, and capacity is their residue (§4).

Ledger-scoped total order. A ledger is one replicated state machine under Byzantine-fault-tolerant consensus among validators its members operate themselves, with no world-global chain, no gas and no fee token, and every quantity in the model computed within one order. A ledger is not a community: a community is a region of the stake graph, one ledger carries as many as its members form, and communities that expect to trade belong on one, which is why there is no bridge (§2, §2.2). Consensus is load-bearing: reservation makes capacity a conserved budget over a shared graph, and a conserved budget needs a serialization point (Remark 4.9).

1.3 The boundaries that remain

Two limits are structural, and neither is acceptance. The limitation usually written here — that nobody can be made to want a medium they cannot hold — describes a bearer instrument; a sale here discharges before it creates (§3.3), so a producer with obligations is paid by ceasing to owe. What binds is the thesis’s antecedent, that enough of what a community buys is available from within it: a property of the community, measurable before founding, which no protocol changes (§10). And zero is absorbing: a community that underwrites nothing at genesis can never leave that state from inside, and §7.4 is its only door. The newcomer’s first uninsured trade is not a third limit but a priced cost: somebody extends a small uninsured risk, it settles, and the stake it writes makes the next trade insured (Remark 4.11).

1.4 Reader’s map

§2 fixes the system model; §3 defines the contract and how it clears; §4 defines capacity and is the core of the design; §5 treats containment; §6 covers default, substitution and arbitration; §7 covers governed constants and re-denomination; §8 states and proves the security results; §9 through §11 take the economic position; and §12 and §13 describe what is built and what is checked.

2 System model

2.1 Community, and what a boundary is

A *community* is a region of the stake graph, not an object: nothing in the ledger names one. There are accounts, underwriters and stakes, and a community is a set of accounts reachable from a set of underwriters. The boundary between two of them is **the absence of a path**.

One ledger therefore carries as many communities as it likes. Two co-operatives sharing one total order stay separate for exactly as long as nobody trades across — no stake edge joins them, so no flow crosses — and join the moment somebody does, with no transition and no ceremony. Measured: two co-ops on one chain, each backed by its own underwriter at 2500, draw 2500 apiece and 5000 together; one settled trade across, and the first underwriter reaches the second co-op.

The boundary that matters is drawn around the *total order*: one replicated ledger, one state, one validator set, every quantity in the model computed within it, and validators that are institutions the ledger’s communities themselves operate. Sharing a ledger is *safer* than not sharing one: a cut is taken over whatever set of accounts one names, so a coalition spanning two co-operatives on one ledger is measured as one set, while nothing measures a coalition spanning two deployments — the scope caveat §8’s guarantees carry.

2.2 One ledger

This design contemplates one ledger. A community does not get its own: one order carries as many regions as its members form, so the case that would call for a second — two communities that want to trade — is the case a shared order already answers.

The rule that holds it is **nothing outside a ledger’s own total order may authorise a transition inside it**, and every discharge in the alphabet already obeys it, being authorised by the party who loses if it is wrong: settlement and cure by the creditor’s signature, an arbitral award by a panel both parties named, capped and time-boxed before the obligation existed.

What the rule refuses is an order honouring proofs drawn from another’s state root. An inclusion proof certifies that foreign validators committed a leaf, and for that to be worth more than a signature the honouring ledger must check it against a root it *knows* to be theirs, which

means tracking a foreign validator set through its changes; anything weaker is a signature by a party who could have signed an ordinary settlement instead. So it is a light client or it is nothing. Measured against a community of six underwriters at 2500: honouring a foreign declared supply yields 6×10^9 where a shared order yields exactly zero, honouring a proof as discharge extracts the community's entire ceiling on *no* local signatures, and a forged discharge recycles to $1000\times$ the attacker's real backing.

Two findings survive the refusal. A genesis file is as free as a key: the design rests on identities being free while stakes are not, so a cut drawn over one order's accounts never measures a coalition that stood up a second. And a forged discharge is cheaper than a default and repeatable, where a defaulter spends their standing once.

A separate deployment is therefore a separate system, with no credit path to it and no protocol for one. A member who wants to participate there trades there — the first trade seats an account and grants nothing — and earns standing there as everybody does.

What bounds one ledger bounds the design, stated as a bound rather than escaped by splitting. §12 measures the binding cost as the per-block re-audit rather than the per-acceptance query: 115 ms on the epoch-advance block over 20,000 accounts with 200 underwriters — the most expensive block there is — against 19 ms for one acceptance. No term grows with the underwriter count, because an obligation's stored hold is a witness the audit verifies rather than a cut it recomputes; what remains grows with the graph. Carrying substantially more than that on one order is open, and splitting does not answer it: severing a stake graph destroys every edge crossing the cut, and two orders becoming one is a founding rather than a migration, so nothing rejoins them.

2.3 Time

Time advances in *epochs* of 86 400 seconds. An epoch number is absolute — the timestamp divided by the constant, identical on every ledger — and a block carries a timestamp from which the ledger derives its target epoch and closes epochs one at a time. What runs *per epoch closed* is what a missed epoch would have to be replayed for: stake decay, the free-allowance and saturation counters, and replay-cache pruning.

What runs *once* at the end of the advance is the sweep: the permissionless cranks that default a matured obligation and forfeit the bonds of a member who has exhausted the write gate for 3 consecutive epochs, followed by bond release. Both are comparisons against the epoch the block ends at, so one pass sees exactly what a pass per epoch would have seen, and a fresh chain climbing thousands of epochs toward wall-clock in one block does not walk the contract book thousands of times. The order of the two is load-bearing: releasing bonds before the forfeiture crank looked would hand back the encumbrance the sanction exists to take. Running the default crank here, instead of leaving it to whoever cares, is what makes §6's substitution automatic: no obligation is ever due and uncranked, so a creditor need not notice their own loss to be made whole.

Epoch boundaries are the only scheduled events: no continuous process, no timer and no off-chain job. Every state change is a transition in the total order or a deterministic consequence of advancing the epoch counter.

2.4 State

The replicated state is, with every amount in it an integer number of minor units (§12):

- **Accounts.** A set of keys, optional guardian configuration for key recovery, cached outstanding debt, how much of it is currently in default, the two support-cascade declarations of §3, operation-bond bookkeeping, and two advisory debt-velocity counters that only a client reads. There is no score, no tier, no attestation and no sponsor.

An account also carries a *status*, which is not an admission ladder and along which nothing is promoted: **active** is the resting state of every account that exists, and the other two are a governance sanction (**suspended**, §7) and a voluntary departure (**exited**, refused while the member owes anything, is in default, holds an unreleased bond, still has a supply declared, or is the last validator). Neither is consulted by Definition 4.5, which is what §4 means by an account having no status.

- **The stake graph.** $w(c, d)$ in minor units, *directed*, keyed by the ordered pair creditor-to-debtor (Definition 4.2); held symmetrically it would say that extending credit raises the lender’s own limit, which the model does not mean.
- **Reservations.** $r(c, d)$, keyed identically, and the committed total per underwriter, together recording the flow held against outstanding insured credit (Definition 4.7). Both halves are state because both are charged (Remark 4.8).
- **Underwriters.** Members who have declared a supply (Definition 4.1), with the amount each declared, every unit of which arrived through a ceremony rather than against standing the community itself conferred (§7). Genesis seeds the first ones; thereafter any member may join or leave the set.
- **Contracts.** Debtor, creditor, outstanding and original amount, maturity, the epoch of acceptance, status, whether the obligation is insured, *exactly the flow it holds* — which arcs and how much on each (Remark 4.8) — and consented arbitration terms if any.
- **Governance.** Proposals, governed constants, validator powers.
- **Chain identity.** Chain id, root salt, replay cache.

There is no reputation structure — no per-relationship evidence, no decayed counter, no reliability estimate, no volume envelope: the stake graph answers directly the question all of those approximated.

2.5 Observables

A member’s client can compute, from public state, its own capacity and who supplies it, its outstanding obligations and their maturities, whether a counterparty’s proposed obligation would be insured, and community-level totals. Precise amounts may be sealed by charter policy to the parties and validators, with others seeing bucketed magnitudes (§12). Capacity is a pure function of committed state, so any observer holding the ledger computes the value any validator does: checkable, not announced.

2.6 Notation

$\mathcal{U}$ is the set of underwriters and $\text{sup}(u)$ the supply u declared. $w(c, d)$ is the directed stake of creditor c on debtor d and $r(c, d)$ the reservation against it, with $\text{free}(c, d) = w(c, d) - r(c, d)$. $\text{Cap}(T)$ is the capacity of an account set T , and $\text{Cap}(x)$ abbreviates $\text{Cap}(\{x\})$. ∂S is the set of stakes with exactly one endpoint in S . V_{base} is the reference denomination, genesis 1 000.

3 The medium: contracts and clearing

3.1 The debt contract

The unit is a debt contract: an obligation of a fixed amount from a debtor to a creditor, with a maturity, in the community’s denomination, created by mutual signature at the point of sale —

the buyer becomes debtor, the seller creditor. It is not a bearer instrument: a creditor cannot sell, endorse or assign it, which keeps a stake conferred by one creditor from being resold to another.

Nor is it frozen. Four things move, each as somebody's discharge rather than anybody's option.

- The **debtor** moves, by transfer or through the support cascade, under the consent rule below.
- The **creditor** moves in exactly one circumstance: default on an insured obligation, where the claim subrogates to the underwriters whose supply carried it (§6.1). Nothing is assigned — the substitution has already made the creditor whole in the same event, and the recovery moves to whoever paid.
- The **maturity** moves only by *extension*, signed by debtor and creditor both, because deferring a debt is the creditor's concession to make. A claim the cascade routes onto a buyer inherits the earlier of its own date and the sale's, so nothing in the alphabet defers a debt without its creditor; an extension past the insured horizon of the claim's acceptance keeps the debt and drops the insurance (§6), on the same two signatures.
- The **outstanding amount**, the **status** and the **flow the obligation holds** move together as it is discharged, cured, defaulted or routed onward, because an insured obligation owes exactly what it holds and the two are one fact (§13).

A contract also records whether it is *insured*: a fact about what it holds, not a label — it is insured exactly while it holds a reservation (Definition 4.7). Decided at acceptance, it can only fall afterwards: when a downward re-denomination rounds its arcs below one minor unit (Theorem 7.3), or in the one place a hold is asked for afresh, the remainder a sale leaves with its seller, if the arcs cannot carry it. A partial payment shrinks the hold by exactly the share paid, on the arcs that held it, and never re-solves it. An obligation gains the community's backing only as a *successor* row somebody signed for, never by the old one changing tier.

3.2 Origination

Acceptance takes both parties' signatures and passes one gate: is the amount within the debtor's capacity, at a maturity inside the insured horizon of its acceptance (§6)? If so the obligation is insured: accepting computes an augmenting flow of exactly the amount into the debtor and holds it (Definition 4.7), so the standing that justified this obligation cannot simultaneously justify another. If not, the creditor may still accept, and the obligation is uninsured: it holds no reservation and the recourse machinery of §6 does not stand behind it. The client says which at the point of decision; a creditor is never uninsured by accident.

3.3 What a sale does, in order

A sale is a discharge first and a claim second: it creates a claim only out of what it could not discharge, and the order is the substance of the medium.

1. **Netting.** Whatever the seller already owes the *buyer* extinguishes first, oldest first — settled, or cured if it had matured. Mutual obligations cancel rather than route, and as an ordinary discharge by the debtor this applies the stake rule of Definition 4.2 exactly as any settlement does.
2. **The cascade.** What netting did not absorb clears the seller's own obligations to third parties, and those of the beneficiaries they have listed, by routing them onto the buyer: a debtor swap, which writes no stake in either direction (Remark 3.3).

3. **The residue.** Only what neither step absorbed becomes a fresh obligation from buyer to seller.

A member who owes is therefore paid by *ceasing to owe*, and a debt is paid here in the usual course by selling to one's creditor: the debt is extinguished and the stake is written, so one act discharges the obligation and raises the seller's standing. §10 takes up what follows, which is most of the answer to why anybody accepts this medium.

3.4 The support cascade

A member may list *beneficiaries*: other members toward whom their own discharge is routed. When a supporter delivers value, the discharge waterfills across the beneficiaries they have listed, in proportion to declared weights, each edge capped at 1 times what that pair has staked in one another (Definition 4.2) and the recursion bounded at 20. The cap is a *stake* and not a settled volume, for Remark 4.3's reason: volume is free to fabricate, so a counter of it would let two cooperating accounts open the drain as wide as they pleased. It has no floor, so a pair with no settled history drains nothing — zero is absorbing here as everywhere else, and escaped the same way, by trading.

This is how a community carries a member through a bad season without anybody making a gift: the supporter's own production discharges the beneficiary's obligations. What the supporter gives up is exactly one thing — the proceeds of their own sale, which would otherwise have cleared their own debts or become a fresh claim on the buyer — and nothing else of theirs is touched, then or later. It carries a member without rebuilding one: a routed discharge writes no stake in either direction (Remark 3.3), so a member cleared this way ends the bad season no better backed than they entered it — which is the whole content of the beneficiary's declaration below, and why it is theirs alone to make.

The cascade records nothing about who contributed what. A co-signer map on each successor contract would be written, cleared on subrogation, rescaled at every re-denomination, checked and committed to the state root while no transition reads it, and ledger state carries what must be enforced; whose production cleared which debt is a question for a client and a social layer, not a field every node replicates forever.

Property 3.1 (A routed claim is never downgraded) *Routing an obligation onto the buyer is a debtor swap, so it owes the same consent rule a transfer does. The affected creditors are discovered inside the transition rather than named by it, and a creditor asked to consent to losing their own recourse would refuse every time, handing every creditor a veto over their debtor's ordinary sales. The rule is therefore enforced as a bound on the amount: **a claim moves only as far as the buyer can carry it insured**, whatever it was before; the part that does not move stays with its original debtor, and the budget it did not absorb falls through to the seller as ordinary new debt. The claim that moves keeps **the earlier of its own date and the sale's**: a debtor swap owes the maturity rule a transfer does, and a claim re-dated to the sale would let two members pass a debt back and forth and push it out of reach for ever, insured throughout, so that no default ever fires.*

Both directions of the bound matter. A claim the buyer can insure is *upgraded* if it was uninsured before, so no signature is owed for it; a claim the buyer cannot insure does not travel at all, which stops a member from shedding an obligation onto a freshly minted key — two signatures and no delivery — and walking away with their own standing restored.

Property 3.2 (Beneficiary consent by moderation) *No member's production is ever routed toward a beneficiary they have not listed, and no member's obligations are ever cleared by a supporter they have not approved. The two declarations sit on opposite members — listing*

beneficiaries is the supporter's, approving supporters the beneficiary's — both are required, and either may be withdrawn.

The two halves answer different questions.

Why the supporter must list. Without their declaration, approval alone would be the attack: naming a wealthy member as one's *supporter* would route their production toward one's own creditors unbidden. Production is directed by the member who produced it and by nobody else.

Why the beneficiary must approve. What a drain costs them is the *standing the obligation would have conferred*. It cannot cost them money, and it is not what keeps strangers out — the cap above does that, at zero for a pair that has never traded whether they approved or not. Routing is a debtor swap the creditor never signs, so by Remark 3.3 it writes no edge: the debt is discharged and the member is exactly as backed afterwards as before, while honouring the same obligation writes that creditor's stake in full. Measured, three loans of 60 cleared by a supporter leave a member at the capacity they started with, where the same three honoured take them from 200 to 380. Capacity is a ceiling and what a drain moves is the room beneath it: the cleared obligation stops reserving what it drew, the release is exact (Remark 4.8), and the member ends with the room they had before they borrowed and the backing they had before the drain — the 200 above. The freed flow is neither awarded to anybody nor lost: the stake edges and the underwriter supply it occupied are available again, to the successor this sale creates on the buyer, to the member's own next purchase, or to any other account those arcs reach, first-come as all capacity is (Remark 4.9).

The beneficiary's declaration is therefore the member choosing **relief now against standing later** — a trade only they can weigh, and one no cap can bound, since the drain cap limits the amount and not the substitution. One smaller thing rides on it: a beneficiary's unused share recurses into their own listing, so accepting a supporter admits that supporter's production into relationships the beneficiary built for their own use.

3.5 Settlement, transfer, and conservation

A contract settles when its outstanding amount reaches zero, by delivery or by mutual discharge. Settlement releases the reservation the obligation held and applies the stake rule of Definition 4.2. A partial payment is at least a 128th of the original unless it closes the row, so an obligation is discharged in at most that many payments and the closing one: a discharge is free, and free writes are bounded by what somebody paid for (§12).

Remark 3.3 (Settlement is the only way the graph grows) *No other transition writes a stake edge. This is what carries Assumption 8.1: every quantity that gates participation — what a member may confer, their operation-bond headroom, the drain cap of the support cascade, the floor on proposing — is read off this graph, so an edge written anywhere but at settlement is standing nobody earned.*

Transfer moves the debtor of a claim: the obligation is assumed by another member, with their signature, subject to their own capacity gate exactly as at origination. The creditor does not change, which is why a claim is not a tradable instrument and a stake conferred by one creditor cannot be resold to another.

A transfer is a discharge of the outgoing debtor, so it must be authorised by whoever loses if it is wrong — the creditor, but only when the claim stops being one the community stands behind. Hence the consent rule: **the creditor signs a transfer exactly when the successor would be uninsured**. Where the successor can carry it insured, the community's recourse

follows the claim to its new debtor, and who that debtor is has become the community's question rather than the creditor's — in this model's terms not a change to consent to. Which case applies is a fact about the graph, not a choice anybody makes.

A transfer also stakes *nothing*: the outgoing debtor did not pay, so Definition 4.2 does not fire, and the graph grows when the successor settles, against the debtor who actually delivered. Otherwise a ring of members passing one accepted claim between them would each collect a stake from a creditor who signed once, and Remark 3.3 would be false.

Invariant 3.4 (Conservation) *At every height, the sum of outstanding amounts over all active contracts equals the sum of cached outstanding debt over all accounts, and no transition creates or destroys obligation except origination (which creates exactly one) and discharge (which reduces exactly one).*

Invariant 3.5 (Reservation conservation) *At every height, $\sum r(a, b)$ equals the total outstanding amount of insured contracts. A reservation exists if and only if there is a live insured obligation behind it.*

Invariant 3.5 is what makes Theorem 8.2 enforced rather than argued: if reservations could drift from obligations, the cut would bound a quantity nobody was tracking.

3.6 Default and cure

An obligation past maturity with a positive outstanding amount is in *default*, and the consequences are asymmetric. For an **insured** obligation the community's recourse machinery applies (§6): *substitution* always, and arbitration if the parties consented to it. For an **uninsured** one none of it applies: the debt stands, the claim stands, and the creditor bears the loss alone (Remark 6.5).

Cure is late discharge: a defaulted obligation may be settled afterwards, which clears the default and applies the stake rule as any settlement does. Default is a state, not a verdict; a member who repays late is not permanently marked, and what marks them is that their counterparties may stake less next time, which the community decides and the protocol does not impose.

3.7 The capacity gate

Invariant 3.6 (Capacity) *No transition leaves any account set S holding insured obligations whose total exceeds $\text{Cap}(S)$.*

This is the only credit limit in the protocol: no per-relationship cap, no per-epoch volume ceiling, no channel with its own floor and no separate rule for newcomers, because Definition 4.5 already gives a newcomer zero and Corollary 8.4 gives a thousand newcomers zero.

4 Standing

A credit community must answer one question about each of its members: how much will it stand behind them. The answer here is a minimum cut — not a figure the ledger stores, but one computed from the stakes and the outstanding obligations that it does.

4.1 Underwriters

Definition 4.1 (Underwriter) *An underwriter is a member who has declared a supply $\text{sup}(u) \in \mathbb{N}$: the amount of the community's credit they are willing to stand behind. $\mathcal{U}$ denotes the set*

of underwriters. Genesis seeds the first underwriters directly; thereafter a supply rises only by an endorsed amendment (§7.4). A member may lower one at any time, never below what they already stand behind, so restating an unchanged declaration is always admissible.

Each half of that rule is load-bearing. Seating by ceremony stops a free key declaring a billion, for a reason no cap drawn from inside the community can supply: a commitment brought from outside cannot be expressed by any quantity the ledger confers, so the ledger asks the community instead. The floor at the current declaration keeps a founder — whose own capacity is zero until somebody backs them — from giving up a standing promise by restating it. An underwriter is not a rank and not a privilege but an accepted liability: a declaration that if the members this underwriter backs fail, this much of the loss is theirs (§6).

The community is not limited by its founders. $\mathcal{U}$ grows as the community endorses new underwriters, so its total underwriting is not fixed at genesis and does not die with the people who started it. Founders are simply the first underwriters.

A declaration may not be made against capacity. The obvious alternative — let a member declare up to their own Cap, since a key nobody backs then has nothing to declare — reads correctly for one member at a time and issues hollow insurance in aggregate: capacity is what the community conferred, so such a declaration becomes a source arc feeding the next member’s capacity, which becomes their declaration. Measured, one founder with a supply of 100 and twelve joiners wash-backing each other, each declaring the maximum that rule allows, reach a declared total of 204,900; each then backs one fresh account, and honest creditors lend those accounts 204,800, every unit of it carrying the ledger’s own *insured* label, with every invariant in §13 satisfied throughout. That the twelve measured *together* can owe exactly 100 is Corollary 8.7, a statement about the twelve; the credit went to the accounts they back, which are not in that set. So a supply has one source, and $\sum_{u \in \mathcal{U}} \text{sup}(u)$ is the community’s insured credit rather than an upper bound on it.

A supply may not be withdrawn out from under live credit. Any member may reduce a declaration, but only down to the flow currently committed through them: the decay floor below, applied to a source arc instead of an edge, because the debt did not shrink when the underwriter changed their mind. Without it withdrawal breaks Theorem 8.2 directly — measured, one of six underwriters leaving a fully drawn community leaves capacity 12,500 against 15,000 outstanding.

An underwriter’s failure is bounded. Supply bounds not only what an underwriter may confer on any one member but the total they may confer on everybody, so the worst an underwriter can do — by folly, by malice, or by backing a coalition of identities they control — is lose their declared supply.

4.2 The stake graph

Definition 4.2 (Stake) $w(c, d) \in \mathbb{N}$ is the stake of creditor c on debtor d : directed, denominated in minor units, updated only by discharge. When d has repaid R of an obligation of A from d to c — at every payment, with R the obligation’s cumulative repayment, so $R = A$ when it settles —

$$w(c, d) \leftarrow \max(w(c, d), \min(R, \Gamma(c))),$$

where $\Gamma(c)$ is what c may confer: $\text{sup}(c)$ if $c \in \mathcal{U}$, and $\text{Cap}(c)$ otherwise. Read over the installment rather than over the obligation, a member who honoured 1,000 in two halves would

hold 500 beside one who paid at once and held 1,000, for the same evidence; the peak over cumulative repayment equals the rule at settlement and counts the installments on the way.

Three properties of Definition 4.2 carry the security argument.

It is directed. That c is willing to carry d says nothing about whether anyone is willing to carry c . Held symmetrically, extending credit would raise the *lender's* own limit — a member could grow their capacity by lending, which is backwards, and which a symmetric graph cannot express its way out of.

It is a peak, not a sum. Running the same cycle of obligations between one pair a thousand times raises w exactly as high as running it once, so a wash loop cannot accumulate however patient its operators are.

It is capped by what the creditor may confer. Nobody can pass on more standing than they hold. Between two accounts that may confer nothing, every settlement stakes $\min(A, 0) = 0$, so no volume of traffic creates an edge at all.

Γ is read live at *settlement*, not at acceptance. For a non-underwriter that is $\text{Cap}(c)$ as Definition 4.5 gives it, net of the credit already standing on c , whose own outstanding insured debt has reserved the very edges the query runs over — so a creditor drawn to their own ceiling passes on nothing when a debtor repays them: what they may confer is what the community currently carries them for, and they are already using it. The two readings differ only when the creditor's own backing moved while the obligation was outstanding, and then the present figure is the true one.

Remark 4.3 (Why stake and not volume) *Letting w record settled volume, the obvious measure of a trading relationship, is unsound: settlement requires two signatures and no delivery, so volume is free to fabricate, and two cooperating accounts can sign arbitrarily large settlements having moved nothing. A stake is not free, because it is capped by a quantity the fabricator had to earn.*

Remark 4.4 (What standing measures) *Because flow runs creditor to debtor, standing is evidence of having owed and paid. Selling earns none of it: measured, 2000 settled sales to twenty well-backed members leave the seller's capacity at zero, while one honoured purchase confers standing at once. Capacity asks whether the community will carry your debt, and the evidence for that is a debt you carried; producing value is different evidence about a different question, and the asymmetry is real.*

It cannot be relaxed within the model. Crediting the selling side by a fraction φ of what the buyer may confer is not a dial: every $\varphi > 0$ reaches the same fixed point, the community ceiling, differing only in rate, because the fraction is applied to a quantity the fraction itself grows and Definition 4.2's peak rule makes it a ratchet. Nor can selling raise some other quantity instead: granting a member whose capacity is 1000 the right to confer 10^6 leaves the newcomer they back at 1000, and a seller with capacity zero granting 10^6 to each of twenty accounts leaves every one of them at zero. A flow network holds exactly two things, source arcs and edge weights; selling can only touch edge weights, and an edge carries only what reaches its tail. To make selling create anything the seller must become a source, and by Assumption 8.1 a source is an external commitment. What is left for a trading history lies outside the flow model: rationing when the ceiling binds, and evidence for the uninsured tier, where the creditor bears the risk and may weigh whatever they like.

4.3 Capacity

Definition 4.5 (Capacity) For a set of accounts T , the gross capacity $\text{Cap}^0(T)$ is the maximum flow into T over the stake graph, where each underwriter $u \notin T$ supplies its whole declared supply:

$$\text{Cap}^0(T) = \text{maxflow}\left(\sum_{u \in \mathcal{U} \setminus T} \text{sup}(u) \longrightarrow T\right).$$

Let $r(c, d)$ be flow reserved against outstanding insured credit, $\text{free}(c, d) = w(c, d) - r(c, d)$, and $\text{com}(u)$ the flow already committed through underwriter u . The residual capacity $\text{Cap}(T)$ is the same maximum taken over what is left:

$$\text{Cap}(T) = \text{maxflow}\left(\sum_{u \in \mathcal{U} \setminus T} (\text{sup}(u) - \text{com}(u)) \longrightarrow T\right).$$

Remark 4.6 (Which one a sentence means) The two answer different questions and are not interchangeable. $\text{Cap}(T)$ is how much more T may borrow, so it is what a transaction consults and what a member is shown, and it reads zero exactly when the ceiling is legitimately full. $\text{Cap}^0(T)$ is how much the community stands behind T altogether, so it is what a bound on outstanding credit must be stated against — comparing credit already drawn against $\text{Cap}(T)$ compares a quantity with itself subtracted out. Every claim below says which it means.

Both subtractions are one statement made on the two halves of a path, and neither is optional (Remark 4.8): free nets what the stake edges are already carrying, $\text{sup} - \text{com}$ what the source arc is. A supply arc read at its full declared value lets every debtor of one underwriter draw the whole supply — the difference between a bound and a suggestion.

Underwriters *inside* T supply nothing to it. This clause is what makes a coalition unable to underwrite itself: appointing your own accomplices as underwriters adds nothing to what your coalition can borrow, because the measurement excludes them along with the rest of the coalition. The set form is the security statement and the singleton $\text{Cap}(x)$ is what a transaction consults; because capacity is a cut, the bound applies to any coalition as a whole, and splitting across identities gains nothing, since stakes internal to a set never cross its own boundary.

4.4 Capacity is generative, not scarce

Definition 4.5 does not divide a fixed pot, and the difference decides whether a community can grow. Backing somebody costs the backer nothing: Γ caps what a member *may* confer, it is not a balance they spend, and a member who backs twenty people has exactly the capacity they started with. Standing also propagates undiminished — a member backed at 2500 may back another at 2500, and so on outward, so capacity reaches strangers along chains of ordinary trade. What is bounded is *simultaneous use*: limits propagate freely, while the credit actually outstanding at any moment must cross a cut and is therefore bounded by Theorem 8.2.

A community's working credit is not the sum of its underwriters' supplies. That sum is an upper bound rather than a measure (Corollary 8.9): it rises whenever anybody declares, and a declaration made against standing the community itself conferred adds nothing the community can carry in aggregate. Working credit rises when somebody accepts a liability the community was not already carrying — an underwriter who can pay from *outside* the set being measured — and §7.4 is the only door to that.

4.5 Reservation

Definition 4.7 (Reservation) Accepting an insured obligation of A for debtor d computes an augmenting flow of exactly A into d and adds the flow it used to r . Settlement or discharge releases it.

Reservation states that a unit of standing backs one obligation at a time, and it is what holds the bound across *independent* creditors. Without it each creditor evaluates a pristine graph and lends against the same edges, and aggregate exposure overshoots the true cut by the number of creditors — measured at $200\times$ with two hundred of them. The overshoot is unbounded, not a constant factor.

Remark 4.8 (What reservation must charge) *Definition 4.7 says the flow it used, and the augmenting path runs from an underwriter’s supply arc through stake edges to the debtor. Both are charged, and an obligation records exactly what it took, so settlement returns exactly that. Charging only the stake edges leaves the supply arc at its full declared value on the next query, so two debtors backed by the same underwriter each draw the whole supply and the overshoot is the number of debtors that underwriter backs — measured with the supply arc uncharged, one supply of 2500 carries 250,000 across a hundred of them. Returning less than was taken is the same error reversed: a release proportional across the arcs incident to the debtor strands the upstream half of every multi-hop path, so settling an obligation destroys capacity nothing was behind. Exactness also disposes of the question of release order, since there is nothing left to choose.*

Remark 4.9 (Why this needs a total order) *The total drawn against a region is order-invariant, but the allocation is not: capacity is first-come-first-served, so which creditor lends depends on the sequence in which acceptances apply. A conserved budget over a shared graph requires a serialization point — the argument that forces a total order for double-spend prevention, arriving by a different route.*

4.6 Decay

Stakes decay per epoch by the governed ratio $977/1\,000$, so standing reflects present backing rather than history. An edge **never decays below its live reservation**: the debt did not shrink because the evidence aged, so the collateral behind it may not either. Free capacity absorbs decay first and is floored at zero, and an edge wholly committed to an outstanding obligation does not decay at all until that obligation clears.

In the unit a member plans in: at the genesis value an unrenewed edge halves in about thirty epochs, keeps 12% after a quarter, 1.5% after half a year and 0.02% after a year; integer decay drops at least one minor unit an epoch below 43, so an edge of 500 is gone in 346. The floor protects only what is still owed, so decay is paid by the member who has repaid and not by the member who is drawn. A community trading on a seasonal rhythm — a harvest a year — is erased between seasons at this ratio, and sizes the governed constant before it sizes anything else (§10.3): at the top of its range a year keeps 69%.

Remark 4.10 (A community that stops trading loses its credit) *A community whose members cease to renew their backing watches its capacity fall to zero — at the genesis ratio, to two ten-thousandths in a year. A community that has stopped trading has no credit, and a system reporting otherwise would be reporting a relationship nobody currently has. What must not happen is that the ability to renew dies out, which is why $\mathcal{U}$ is open to any member rather than fixed at genesis.*

4.7 Insured and uninsured credit

Capacity bounds what the community *underwrites*, not what a member may choose to risk. Within the debtor’s capacity an obligation is **insured**: it reserves flow, and the recourse machinery of §6 stands behind it. Beyond it a creditor may still lend, and the obligation is **uninsured**: it reserves nothing, triggers no community recourse on default, and the creditor bears it alone.

This dissolves the bootstrap. Were capacity a permission, a community whose accounts all begin at zero would deadlock, since the first trade could never happen; instead first trades are uninsured, they settle, they create stakes, and capacity is their residue. Uninsured credit still stakes under Definition 4.2, so an uninsured loan from an account that may confer nothing confers nothing.

Capacity then grows *insured across creditors and uninsured within one*. The cut sums every honoured relationship, and a loan from any creditor is insured against the whole of it, so each settlement that writes a new creditor's edge raises what the next loan may be insured for. Measured: one uninsured loan of 50, then eight loans inside capacity from three creditors in rotation, take a member through 100, 200, 350, 650, 1,200, 2,000 and 2,650 to 3,000 — the seed's whole reach through those three — with 7,200 lent insured on 50 lent uninsured. Within a single relationship the stake is a peak, so a limit at one creditor rises only by a loan above it, and reservation being whole, that loan is uninsured in full: the marginal risk sits with the party holding the information about the debtor, and the community insures what has been demonstrated.

Remark 4.11 (The friction this leaves) *An unconditional starter allowance would make onboarding frictionless, and is exactly the hole that makes identities worth minting: N free keys would carry N allowances. Refusing it moves a real cost onto the newcomer — somebody must take a first, uninsured risk on them, and no mechanism here supplies that willingness. That is how a trading relationship has always begun, a shopkeeper letting a new neighbour run a tab, and this design's part is to make the risk small, bounded and legible: the amount is the creditor's own choice; the exposure is exactly that amount and never more (Cor. 8.10); the client states at the point of decision that the obligation is uninsured (§3); and the bond that same counterparty pays is what seats the account (§12). The risk is what writes the stake, and a mechanism that removed it would remove what capacity is made of.*

4.8 What an account is

An account is a set of keys and the bookkeeping that hangs off them. It carries no tier, no attestation, no sponsor and no score, and in particular no *admission* status: there is no rung to be on and none to be promoted from. The status it does carry is the governance sanction and the member's own departure (§2), and neither is consulted by Definition 4.5 — the graph answers the same for a suspended account as for an active one, and what suspension revokes is origination, not standing.

It is seated by its first *bonded trade* — an acceptance or a sale naming its key as a party, signed by that key — and requires no approval, because creation grants nothing: an account with no incident stakes has capacity zero by Definition 4.5 and confers nothing by Definition 4.2. Seating by trade rather than by a creation transition is a cost argument, not a permission one (§12): a row that only ever appears beside a write somebody with standing paid for is bounded by that standing, while an unbillable creation is bounded by nothing and cannot be given a quota that is not also a censorship lever. The counterparty who carries that first bond is the one already carrying the first uninsured risk (§3); nobody else is asked, and nobody may refuse on their behalf. The row is written *last*, after every check that could refuse the trade, so a refused transaction leaves nothing behind.

The row is a stock, and its price is one: the seat holds a bond unit of the counterparty's reach on a layer credit never reads, for as long as the row exists (Property 12.2). A bond is reserved and returned, so it prices how fast rows appear and not how many; no transition returns a seat, because no transition retires a row. What does retire one is the epoch sweep, a year after the seating, when the row holds nothing, owes nothing and is named by nothing — no stake in either direction, no obligation in any status, no bond, default, supply, vote, pending rotation, panel, proposal or guardian roll, and no live row it sponsored — and then

the seat and the key come back. Each of those is something only the member, a counterparty or a ceremony can put there, so the rule is not a lever anybody can pull against another, and a row that is empty is not a stock. The last condition is the sweep's own: a seat names its sponsor for the life of the row it bought, so a sponsor whose own trade has gone quiet stays until the rows it seated are gone, and goes at the boundary after the last of them.

What an account accumulates is not a property of itself but of the community around it. Standing is held in the stakes others place on it; no account owns its own capacity, and none can raise it alone.

5 Stability and containment

A credit community fails in one of two ways: it underwrites more than it can absorb, or one member's failure propagates until many fail. The mechanism of §4 does most of the work against both.

5.1 Aggregate exposure is bounded by construction

The total insured credit a community can extend is not a quantity anybody sets. By Corollary 8.9 it is bounded by the sum of the supplies its underwriters have declared, a figure the community explicitly committed to. That sum is a bound and not a measure: what a community can carry is what its underwriters can pay from outside the set being measured, and the two diverge sharply once members declare against standing the community itself conferred (§4). Nor can the community over-issue by increments: every unit of insured credit is backed by a reservation on a specific chain of stakes running to an underwriter, and when those are exhausted the next acceptance is uninsured — visibly, at the point of decision, to the creditor who would bear it. No aggregate silently inflates.

5.2 No separate macroprudential brake

This design has no brake: no system-wide multiplier that throttles new credit when measured exposure rises. A brake is an exception to the credit rule, applied on a different measurement on a different schedule — a second answer to "how much may this member owe", the question this design exists to have exactly one answer to — and as a global multiplier it reprices every member's capacity for conditions none of them created, squeezing hardest the members with the least to do with the exposure that triggered it. What a brake was for is achieved structurally: exposure cannot rise without reservations rising, reservations cannot rise beyond the cut, and stakes decay continuously unless renewed, so a community whose trade slows sees its aggregate capacity fall without anybody measuring anything.

5.3 Where a loss lands

A default on an insured obligation is absorbed by the underwriters whose supply carried it, and by nobody else (§6.1). The loss does *not* travel along the chain of stakes the reservation ran through: intermediate stakes on a reserved path are conduits of evidence, not of liability. A stake is placed mechanically at settlement — a member who honours a debt thereby records that their creditor stood behind them — so liability along the path would be an externality nobody signed, attached to the act of trading honestly, chilling exactly the trade that builds the graph and most for the members with the most trade behind them. The sanction therefore lives where the consent lives: at the source arc, whose owner declared a supply and thereby consented to inherit debts up to it (Property 6.1). A member in the middle of a path loses nothing when a debtor two hops away fails, and owes nothing on their account.

5.4 What the cascade couples, and what it does not

The support cascade (§3) is the one mechanism that couples members’ fortunes, and **it couples production, not failure**. When a supporter sells, the proceeds discharge a listed beneficiary’s obligations rather than the supporter’s own. What the supporter gives up is that clearing — the debts of theirs that would otherwise have closed, or the claim on the buyer they would otherwise have held — and nothing attaches to them afterwards. If the beneficiary later fails, the loss falls where §5.3 says every loss falls: on the underwriters whose supply carried the obligation, or on the creditor alone if it was uninsured. **No mechanism in this ledger makes a beneficiary’s failure a supporter’s**, and there is no record for one to be built on: the cascade writes no attribution at all (§3).

Three bounds sit on the coupling that does exist. The recursion is capped at depth 20. Each edge is capped at 1 times what that pair has staked in one another — zero for a pair that has never settled anything, so a stranger who lists you drains nothing whether you approved them or not. And both declarations are required (Property 3.2), so no member is drawn in without having said so. The residual exposure is therefore explicit and belongs to the *supporter*: the set of beneficiaries they have *listed*, visible in their own client, revocable by them, and bounded above by what they choose to sell, while what the beneficiary gives up by approving is standing, not money (Property 3.2). This is a smaller guarantee than a system-wide containment bound: coupling that members opt into cannot be bounded by the protocol without overriding the choice they made.

5.5 Decay as the withdrawal mechanism

Decay (§4) is how a community withdraws backing without any member having to act: standing given once would otherwise be given for ever, and a relationship nobody currently has would go on being underwritten. Its floor at the live reservation is what stops the same mechanism dissolving collateral out from under a debt still owed.

6 Recourse

What happens when an obligation fails depends on whether the community stood behind it. An insured obligation is settled by *substitution*: the underwriters whose supply carried it become the creditor’s debtors, and the defaulter’s debt moves to them. Beside it sit two optional instruments, an arbitration panel the parties consented to in advance and the forfeiture of a member’s operation bonds for abusing the write surface. An uninsured obligation has no recourse machinery at all (§6.4). None of it creates money.

6.1 Substitution: what an underwriter actually pays

Corollary 8.9 bounds an underwriter’s loss; this section settles it. With no external redemption there is no asset to hand over, so one constraint shapes the answer:

In a closed system, the only way anyone can pay is to become a debtor. “You lose x ” can only mean “you now owe x of goods and services.”

A default on an insured obligation is one event with two movements. In *substitution*, the creditor’s claim lands on the underwriters, split by exactly what each one’s supply arc carried — a lookup, since the obligation records which arcs it drew and how much on each (§4) — and each underwriter becomes an ordinary debtor of the creditor, discharging the way anyone discharges anything. In *subrogation*, the defaulter’s debt runs to the underwriters instead of to the creditor, so a defaulter who cures repays the underwriter, and the creditor, already whole, is not paid twice.

Property 6.1 (Consent through declaration) *Every discharge is authorised by the party who loses if it is wrong. Here that party is the underwriter, who authorised it by declaring a supply: a declaration is not a number but a signed, standing consent to inherit the debts of those the community's stakes reach through the declarer, up to the declared amount.*

The claim keeps the flow it held, with three consequences. The defaulter's standing stays consumed, so paying a loss does not restore the member who caused it. The underwriter's supply stays drawn until the defaulter repays them, so unpaid absorbed losses throttle the declaration and §4's withdrawal floor stops them walking out from under it. And the reservation totals do not move: the pieces partition the original reservation arc by arc.

A consented panel stays with the claim. Its terms record the parties and the amount they were consented for, so the change of creditor does not reach them: the award is still the buyer's remedy against the seller for non-delivery, minted between them, bounded by the original amount, and never against an underwriter who was never asked. Dropping the panel with the party would close the remedy in the one case arbitration exists for, a buyer who withholds payment for non-delivery and defaults.

The substituted leg is *uninsured*. Insuring it would draw on what others have put behind that underwriter — the loss walking one hop further out, the path contagion §5 refuses. If the underwriter defaults on it too, the creditor bears the loss, visibly and attributably; without an external asset the system cannot do better and must not pretend to. What substitution buys is that a supply declared by somebody who cannot honour it fails in public.

Remark 6.2 (An underwriter's loss is bounded; their time is not) *Committed flow is released by the defaulter repaying and by nothing else — not by the underwriter satisfying the creditor, and not by time. An underwriter who has honoured a substituted obligation in full therefore stays floored at the committed flow, unable to withdraw the supply, exit the role or put a date on either, for as long as the defaulter declines to cure. That is the throttle: releasing on a timer would let an underwriter absorb a loss, wait, and insure the same amount over again. What is bounded is the amount (Corollary 8.9); the duration of a default is the defaulter's. Extend produces the same lock with no default at all, on the signatures of the debtor and creditor alone — but only within the insured horizon of the claim's acceptance, 365 epochs at genesis and governed (§7): an acceptance or an extension past it books or leaves the claim uninsured, the flow returns, and an underwriter's exposure in time on a performing claim is bounded by a horizon the electorate — the underwriters — set in advance. The horizon is measured from the acceptance, which a transfer and a routed successor inherit, never from now, so no chain of extensions rolls it; keeping a claim insured longer is settling and re-accepting it against the current cut.*

6.2 Consented arbitration

Parties may attach arbitration terms at acceptance: a panel of named arbiters, a quorum, a window and an award ceiling, pinned there and immutable — a channel cannot be opened, widened or repointed afterwards by either party. Arbiters within the window each attest an amount; at quorum the award is the *median* of the attestations, minted once as an ordinary obligation from the original **creditor** to the original **debtor**, the buyer's remedy for non-delivery: an obligation records that the debtor received value and owes for it, and the dispute an award settles is whether the creditor delivered it. The award is bounded three ways, all consented at acceptance — the median, the agreed ceiling, and the original amount.

The remedy is available in *any* state of the original obligation, settled included, and after substitution, when the row has become an underwriter's claim (§6.1): a buyer who has paid in full and received nothing is the case it exists for, and gating it on the obligation still being open would close the channel the moment it falls due.

Property 6.3 (Median capture resistance) *With a quorum of q arbiters, an award is moved only by a coalition holding more than $\lfloor q/2 \rfloor$ panel seats. A minority, however extreme its attestations, cannot move the median.*

Property 6.4 (An award is bounded and uninsured) *An award is minted at most once per contract, never exceeds the original amount or the agreed ceiling, and is **uninsured**: it reserves no flow, draws on no underwriter, and substitutes nothing if it is never honoured — it is worth exactly what §3 says any uninsured claim is worth. What it costs the loser is their write channel, since an operation bond is measured against headroom net of what the member owes (Property 12.1), and it nets against any future trade the loser does with the winner, the ordinary way anything is paid in a mutual-credit community.*

An award is therefore worth less here than in a court. The alternative would draw on the community’s underwriters to satisfy a private dispute between two members: capacity bounds what the community underwrites, and nobody underwrote the quarrel. The panel size is capped at 16, and arbiters may not be either party.

6.3 Netting rings of defaults

A sale nets before it creates: whatever the seller already owes the buyer extinguishes first (§3.3, step 1). That is a cycle of length two, and nothing in the argument depends on the length: in a ring of A owing B owing C owing A , netting the smallest of the three relieves each party of exactly as much debt as claim.

The epoch sweep does it with no signature, because every hop is in *default*: every party is already due, so nobody is paid early, nobody loses time value, and each is relieved of a debt they did not pay against a claim they could not collect. Netting a live obligation would hand its creditor an early payment they never agreed to take, so the default condition is a rule rather than a heuristic, and a claim under an open arbitration window belongs to the panel rather than to the sweep.

Each hop is an ordinary discharge and earns standing accordingly, capped as always by what the creditor may confer: the debtor gave up a claim of their own worth exactly what they were relieved of. Where a hop was insured, netting *cures* it and the reservation follows the debt down, which a default does not do on its own.

Substitution breaks a ring of insured defaults, and that is not a defect: an insured claim moves onto the underwriters at default (§6.1), so its creditor becomes an underwriter and the cycle is a cycle no longer. What the sweep finds is therefore mostly rings of *uninsured* defaults, where nothing else was going to help: no underwriter stands behind them and the creditors bear them alone.

Zero-priced and permissionless, it is bounded like every other crank: at most 8 hops to a ring and 64 rings to a sweep. Every ring netted closes at least one hop and so destroys its own precondition, so a sweep cannot find the same ring twice, and what is not reached this epoch is reached the next.

6.4 The uninsured tier, and why there is no pool

The layer that suggests itself is a funded loss pool: members take a covenant — a per-event contribution cap, a committed balance and a ceiling on total commitment — and a creditor covered by it claims against a defaulted *uninsured* obligation. Assumption 8.1 rules it out.

A covenant is funded by nothing the ledger can hold. A commitment reserves no flow and is checked only against a ceiling the member declares for themselves, so it is a signature in exactly the sense the assumption means: measured, a member holding 1.0 of standing pledges 10^9 , is displayed to every creditor as coverage, and mints 399.9998 of a 400 claim while an honest

member’s 500 mints nothing — an uninsured obligation the pledger never honours. Capping the pledge does not repair it: a cap is checked once and not maintained, so a member may pledge at their ceiling and then borrow the same capacity away before any claim arrives. The only way to make a pledge funded is to *reserve* flow, which is the insured tier — reserving against the claim actually at risk, at the moment it is taken, bounded by the cut, subrogated when it fails. A pool cannot be made sound without becoming substitution, and the measurement in §8 is the same fact from the other side: a mutual pool funded from inside re-labels the seed instead of adding to it.

Remark 6.5 (The uninsured tier is borne alone, and that is what earns standing) *A creditor who lends beyond a debtor’s capacity bears the loss — §3’s asymmetry, unqualified. Settlement is the only way the stake graph grows (Remark 3.3), so a creditor’s own risk on a first trade is what confers standing on a stranger. Socialising it would make standing manufacturable: a ring covenanting for one another could trade inside itself, default, and extract every honest pledge for the price of restitution it never honours.*

Mutual aid needs no transition. A member who chooses to carry a neighbour’s loss issues the injured creditor an obligation, or nets it against their own trade, at the moment the loss is real — funded when it is used, targeted at the loss that happened, and signed by somebody with the capacity to mean it. What the pool added was a number the ledger could not keep.

6.5 Operation bonds

Writes are feeless: no gas, no fee token, no revenue to anyone (Property 9.1). Spam is priced instead by *bonds*: a transition beyond a member’s free allowance of 32 per epoch encumbers headroom equal to the bond unit $0.02 \cdot V_{\text{base}}$, which returns after 1 epoch unless forfeited. The headroom it is measured against is not the declared supply but what the community’s external seed reaches the member for (Property 12.1).

A bond is reserved headroom, never a balance: nothing in it is owed to anyone and every entry returns on schedule. Only sustained exhaustion — being denied by the gate in 3 consecutive epochs without returning under the ceiling — makes bonds forfeitable, and a forfeited bond is recorded against the member rather than paid to anyone.

Property 6.6 (A forfeiture is a reservation that never returns) *Forfeiture removes the encumbered headroom permanently: the amount is recorded against the member and subtracted from their bond headroom from then on. Nothing is minted, nobody is owed it, and no position anywhere improves, so the sanction is enforceable by arithmetic and still collects no rent (Property 9.1). The way back is the model’s own: a member whose standing grows beyond what was taken has a write channel again.*

Two exemptions keep the sanction from being absorbing. The free allowance still applies, and the recovery path — discharge, cure, transfer, exit, and reducing a supply — is priced at zero, so a member who has lost its headroom can still pay what it owes, provided somebody can sign for the payee. A discharge needs the creditor’s signature, and a creditor who has lost their keys or stopped answering would otherwise make their debtor a permanent defaulter — an uninsured default substitutes nothing, and an open default holds the allowance and the exit. So a threshold of the creditor’s *guardians* may sign a discharge in the creditor’s stead: they already hold the power to rotate the creditor’s key and sign as them, so this adds no trust the creditor had not placed. A creditor who registered no guardians and answers nothing leaves a debt nobody can discharge, a fact about that creditor the wallet states when a counterparty has none. The same exemption keeps the *governance* sanction from being absorbing: a suspended account’s headroom is zero by status rather than by exhaustion, and a suspended underwriter

may still lower what they stand behind. A refusal for want of a budget that was never charged is reported as its own condition, because the remedies differ — one waits for a bond to release and the other cannot — and because the exhaustion refusal is the evidence the forfeiture crank counts, which must not accumulate against a member for asking.

The allowance is not what carries a newcomer: it goes only to accounts with something to lose (Property 12.1), or N free keys would carry N allowances. What carries a newcomer is the counterparty, whose bonded first trade seats the row and holds a seat on their reach for its life (Remark 4.11, Property 12.2).

7 Governance and indexation

7.1 Three classes of constant

Genesis data. Fixed when the community is created and never amendable: the chain id, the root salt, and the validator floor — how few validators any removal may leave standing, 4 for a real chain and 1 for the harness one. Governing the first two would defeat what they establish: a chain id that could change would stop separating one network’s signatures from another’s, and a salt that could change would stop protecting the preimages already committed under it. The floor is the ceremony’s because only the ceremony knows which kind of chain it is founding, and the ledger holds it afterwards, because a floor the ceremony alone held binds the founding and nothing after it: a four becomes a three on one validator’s free exit, and a one after three, on no vote at all.

Two *sets* are also seeded at genesis, and neither is frozen: somebody has to be first. The validator set’s initial composition is genesis data, but voting power is amendable at runtime by proposal, which is how a community replaces an institution that has left or misbehaved without starting a new chain. The initial underwriters are genesis data in the same sense: a supply rises thereafter only by an endorsed amendment (§7.4), and any member may lower one. A declaration confers nothing on the declarer and a coalition cannot underwrite itself (Corollary 8.7); what it does is accept liability.

Governed constants. Amendable within static safe ranges by proposal and assent. There are six: the client-advisory risk midpoint, the visibility policy, the operation-bond fraction, the stake decay ratio, the seed-amendment rate β (§7.4), and the insured horizon — how far past its acceptance a claim may mature and still hold a reservation (§6). Every one is *dimensionless*, which is what lets a static safe-range table stay exact across re-denomination: a denomination-valued constant would drift out of its own constitutional range the first time the unit was rescaled, while the range still read the same numbers.

Frozen constants. Amendable by nothing. The adoption threshold Θ_{adopt} , genesis 0.5, is the mass an assenting coalition needs to enact an amendment — including one targeting the threshold itself — and Θ_{val} , genesis 0.6666666666666666 — two thirds, to the last bit a double carries — is the mass it needs to change who orders the ledger.

7.2 Amendments and adoption

Assenting mass. The weight of an assenting coalition is its share of the community’s *external* seed:

$$\text{mass}(A) = \frac{\sum_{u \in A} \text{ext}(u)}{\sum_u \text{ext}(u)},$$

where $\text{ext}(u)$ is the part of u ’s declared supply that arrived through a ceremony — genesis, or a seed amendment the community endorsed. Every other candidate was measured and

failed: a headcount is a count of free signatures, and a cut over the assenters counts internal declarations at face value and measures who the seed *reaches* rather than who put it up — a founder who declares the whole seed and backs one member for all of it has capacity zero, and that member alone carries the vote. The external seed is the one quantity in this system that a signature cannot manufacture, Assumption 8.1 applied to votes instead of to credit. Members may propose — the floor is only that the proposer has something to lose — and an assent from outside the electorate is refused rather than recorded at zero. Mass is monotone in A by construction, so assenting can only ever help.

Members assent; when assenting mass reaches the threshold for its kind, the proposal enacts. There are six kinds and one measure, and two thresholds:

- a **parameter change**, naming one governed constant and a value;
- a **re-denomination**, the atomic change of unit below;
- **suspension** and **reinstatement** of a member — the only sanction the community votes on, as against the arithmetic one of Property 6.6. It revokes *origination*, not standing: a suspended member may still discharge what they owe, be paid, net a sale against debts they already carry, extend an existing obligation with the creditor’s own signature, accept relief from a supporter, lower a declared supply, and register guardians or rotate keys — none of those is a new obligation, shutting the first would be a debtors’ prison, closing the supply door would make the sanctioned a permanent involuntary insurer, and closing the last would let a sanction about conduct strand a member with a stolen key. One proposal stays open to a suspended member, to write and to vote on: reinstatement, and nothing else, so that an electorate cannot sanction itself out of existence (below). What suspension does close, beyond origination, is the member’s own write budget: a non-active account confers as before but has no headroom of its own, so what remains open to it is paid for by whoever else signs;
- a **validator power**, which sets one member’s voting power and removes them at zero;
- a **seed amendment** (§7.4).

Changing who orders the ledger takes two thirds. A validator power, and a suspension or reinstatement whose target currently holds voting power, need Θ_{val} rather than Θ_{adopt} . The distinction is recoverability: a constant moved too far is moved back through the same door, and every value inside a constitutional range is one the ledger keeps working at, while a coalition that holds the validator set decides which blocks exist, including the ones that would undo it. At one threshold for every kind, a coalition holding half the seed removes every other validator down to the ledger’s floor and suspends whoever objects, alone and in one epoch. Two thirds is the bar the consensus below already uses: a quorum is $2f + 1$ of $3f + 1$, so a set changeable by less than two thirds is changeable by a minority the protocol already tolerates as faulty. Whether the higher bar applies is read off the target’s current voting power rather than off the proposal’s kind, because suspension removes a validator too and a rule keyed on the word would leave the same door open one name over; the reading is a function of committed state, so every node agrees which bar applies.

The first two kinds carry a cooldown of 90 epochs — per constant for a parameter change, one for re-denomination — which bounds how fast a dial can be walked. It is checked before the enacting assent is recorded, because the state machine has no rollback: a refusal that had already banked the assent would let a later one enact on a support set including somebody who had been told they were refused.

One measure decides all six because there is almost nothing to leave on a lighter tier. A bond fraction at its ceiling is censorship by arithmetic; a stake decay at either end of its range

is a credit freeze enacted as a rounding rule; an insured horizon at its floor is the same freeze enacted as a term; β governs how fast the electorate itself may grow; and the visibility policy, flipped, unseals every amount in a community that had chosen to seal them. Five of the six governed constants take something from members who did not vote; the sixth is a client-side score constant no consensus rule reads, and a second measure for that one constant would be two rules for one act, in exchange for keeping alive a measure the paragraph above rules out.

Property 7.1 (Safe-range soundness) *A value outside a constant’s safe range cannot be enacted, whatever mass supports it. The state machine refuses it at application, so the range is a property of the ledger rather than of the proposers’ restraint.*

The ranges are bounded on *both* sides, and for the bond fraction and the decay ratio the ceiling is the load-bearing half, for the reasons above: a hostile amendment must be loud and attributable rather than economic.

7.3 The constitutional freeze

Property 7.2 (The door cannot amend its own guard) *No governed constant is Θ_{adopt} , and no amendment can write to it. The freeze is enforced by omission: the key space of amendable constants has no variant naming it, so no proposal can be constructed that targets it.*

Without this, a coalition holding the threshold could lower it and then adopt anything at the lowered bar, so the guarantee that amendment requires broad assent would be exactly one amendment deep.

7.4 Seed amendments

A supply is written by a ceremony and by nothing else (§4), which keeps a free identity from declaring a billion — and, with genesis the only ceremony, would close the ledger’s external-commitment door for everybody: a founder could never raise their supply, a newcomer bringing backing from outside would have no transition to arrive through, and a group forming on an established ledger in year three could never have underwriters at all. Genesis fixes two sets, the validator half is amendable at runtime, and the credit half must be too.

No set of signatures can prove externality (Assumption 8.1), so there is nothing to verify: genesis was never a verification but a *ceremony* the ledger recorded, humans checking each other out of band with the file as the record. The repair is to make the ceremony repeatable, and a *seed amendment* is that: an ordinary proposal, carried by ordinary assent, whose effect is to add its amount to the proposer’s declared supply outside the capacity cap. Three properties make it safe to have at all.

The proposer is the beneficiary, structurally. A supply is a standing consent to inherit the debts of those the community’s stakes reach through the underwriter (§6), so an amendment naming somebody *else* would volunteer a member to underwrite, and every discharge here is authorised by the party who loses if it is wrong. The proposal kind carries no beneficiary field: its beneficiary is its author, so the consent is the author’s own signature and cannot be omitted. The author may not assent their own amendment, because assenting mass is a share of the external seed and an amendment adds external seed to its author: without the exclusion, an underwriter would vote their own raise through on the strength of the very quantity the raise enlarges, each one carrying the next more easily.

The rate is bounded, and bounded on the right base. Amendments enacted in one epoch may add at most β times the seed the epoch opened with, and the headroom does not carry over between epochs. Every unit of every supply arrived through a ceremony (§4), so the roll *is* that seed; a rule that admitted declarations against conferred capacity would give the base an inflatable second half and turn the same multiple into amendment headroom (§4 measures it).

Zero stays absorbing. A community that underwrote nothing at genesis has an external seed of zero, and β times zero is zero, so no amendment can ever be admitted — and it has no assenting denominator either, since the denominator *is* the external seed, so no proposal of any kind can be enacted. Assumption 8.1 is untouched: a repeatable ceremony still needs a seed to be a fraction of.

The bound changes the *shape* of the residual risk without removing it. A coalition holding Θ_{adopt} of the external seed can still assent to a commitment that does not exist; no rule prevents a community from lying to itself, and none ever will. The bound turns that from an explosion into a slow, public, attributable leak — measured at $4.3\times$ over thirty epochs with β pinned to its constitutional ceiling and every amendment the community could pass being passed — during which the only parties at risk are lenders, who must still choose to deliver real goods against the inflated figure, and who can simply stop. The same humans were trusted absolutely at genesis; the mechanism adds no trust assumption, it makes the existing one repeatable and auditable.

What making the seed the electorate exposes. Three things, recorded rather than mitigated. Ordinary members have no vote: that is the decision itself — the premise given up is that credit trust implies a governance mandate, and the scene above is why there is no sound capacity-based member vote to keep instead; a community that wants to widen its electorate does it the way it grants credit, by endorsing a real commitment. The electorate is a specific, listed set of accounts, so the constitution’s soft underbelly is the *guardian sets of the external underwriters*: key recovery (§12) is opt-in, self-declared and veto-windowed, but a threshold of an underwriter’s own guardians seizes a vote along with an account, and narrowing that would be a rule about who may be trusted with a key, which is not a rule this design knows how to write. And suspension silences a voter without removing their weight from the seed, so a coalition at the ordinary bar could suspend everyone else and withdraw, leaving an electorate in which nothing could ever be enacted again — the argument that a constant is recoverable through its door assumes the electorate survives its own sanctions. So the sanctioned keep one vote, on reinstatement: a majority of the seed cannot be held suspended, which is what a majority of the seed means, and a minority still cannot reinstate itself.

Finally, an amendment endorses something concrete: a substitution liability (§6), enforceable, and failing in public if it was never real. Amendments without substitution would endorse nothing.

7.5 Re-denomination

Prices in the community unit drift against the outside economy. The response is *re-denomination*: an atomic, governed change of unit by a factor π , bounded per event by $|\ln \pi| \leq 0.5$.

Theorem 7.3 (Exactness up to the minor unit) *Re-denomination by π multiplies every denomination-valued quantity — outstanding amounts, stakes, reservations, bond encumbrances, the reference denomination — by π simultaneously, and leaves every dimensionless quantity unchanged. Consequently every member’s capacity and every ratio of capacity to debt are preserved exactly in the reals.*

Proof. Capacity is a maximum flow over $\text{free} = w - r$. Scaling every w and r by π scales every cut by π , hence the minimum cut by π , hence capacity by π . Outstanding amounts scale identically, so each comparison of amount against capacity has both sides scaled by the same positive factor and its outcome is unchanged. $\square$

The theorem is why stakes and reservations must be in the rescale set: a re-denomination that moved the constants and the debts but not the stake graph would silently reprice every credit limit in the community while appearing to be a change of unit.

What “exactly” cannot mean at the minor unit. The capacity path is integer — every stake, supply and reserved arc is a whole number of minor units — and π is rational rather than integral, so the products are not. Every one is rounded *down*, the only safe direction, since rounding a stake up would let a reservation claim standing that nothing carries. Three consequences follow, and the implementation states them where the rounding happens.

An insured obligation *owes exactly what it holds*: its amount is re-derived from its re-denominated arcs rather than scaled on its own, so it rounds down by at most one minor unit per arc. Scaling the amount independently would put $\lfloor \sum \rfloor$ against $\sum \lfloor \cdot \rfloor$, and §13’s reservation invariant asks for equality, not closeness.

The reservations are a *flow*, not a list, and one supply arc may carry several paths. The supply side therefore rounds down once where the stake arcs beneath it round down separately, so after the flooring each debtor’s supply claim is brought back to what their arcs can still deliver — per debtor, because it is the union of a debtor’s holds that is an augmentation, and a subrogated claim holds a share of each arc rather than a path of its own (§6).

An obligation smaller than one minor unit holds nothing, and an obligation that holds nothing is not insured, so a downward re-denomination can move a sub-minor claim into the uninsured tier. The debt itself is untouched; what it loses is a reservation that had already rounded to zero.

8 Security

The security of a mutual-credit system is the question of how much an adversary can make the community underwrite. Here the question has a closed-form answer, because capacity is a cut and cuts compose.

Throughout, S denotes a set of accounts and ∂S the set of stakes running *into* S from outside it. All statements are about a single *membership* — one account set, one total order, one underwriter set, one cut. §2 explains why nothing is ever computed across two orders, and §2.2 why there is one.

8.1 What cannot be manufactured

Every result in this section applies one statement repeatedly.

Assumption 8.1 (The free-signature bound) *Every ledger event is a signature, and signatures are free. No quantity computed from ledger events alone can therefore seed insured credit: the seed must be a commitment by a party with something to lose outside the ledger.*

Identities are free, settled volume is free, communities are free — a genesis file costs nothing — and a levy on a free quantity is free. The only unforgeable object in the system is a declared liability, and capacity is the arithmetic of tracing how far one reaches. Measured against the independent max-flow: a mutual pool funded by a levy on every settlement, started with no underwriters, accumulates *zero* over 10,000 settled trades, because nothing may be conferred, so nothing is staked, so the levy is a percentage of nothing. Seeded, the pool does accumulate

real stake — and the members measured *together with* it still owe exactly the seed: a pool funded from inside re-labels the seed instead of adding to it, which is why the ledger has no loss pool (§6.4). A community that underwrites nothing at genesis can trade indefinitely on uninsured credit and can never leave that state, since a supply is seated only by a ceremony and a ceremony needs a seed to be a fraction of.

8.2 The cut bound

Theorem 8.2 (Cut bound) *Let $\text{drawn}(S)$ be the total insured credit outstanding to the members of S that is sourced by underwriters outside S . Then*

$$\text{drawn}(S) \leq \text{Cap}^0(S) \leq \sum_{(a,b) \in \partial S} w(a,b) + \sum_{u \in \mathcal{U} \setminus S} \text{sup}(u).$$

Proof. By Definition 4.7 every insured obligation to a member of S holds an augmenting flow into S , and distinct obligations hold disjoint flow, since reservation subtracts from free before the next augmentation is computed. The total held is therefore a feasible flow into S over the gross graph, bounded by the maximum such flow, which is $\text{Cap}^0(S)$. The second inequality is max-flow/min-cut: ∂S together with the supply arcs of underwriters outside S forms a cut, so the maximum flow is at most its capacity. $\square$

Remark 8.3 (Why the gross reading, and why flow sourced outside S) *Both qualifications are load-bearing, and each fixes a different way the statement is false without it. Against $\text{Cap}(S)$ the theorem fails for any drawn debtor: the residual reading nets out exactly the credit the claim is about, so a member who has borrowed to their ceiling has $\text{Cap}(S) = 0$ and $\text{drawn}(S) > 0$. And drawn counts flow sourced outside S because Cap^0 does: an underwriter inside S supplies S nothing in the measurement, so credit it insures for its own set is neither bounded by this nor claimed to be. This is what the implementation checks after every committed block (§13, invariant 1).*

The proof uses *disjoint flow*, which binds the implementation as much as the argument: an augmenting path runs from a supply arc through stake edges, and a reservation must charge both (Remark 4.8). Charging only the edges leaves the theorem true of Cap while the system reaches states it excludes, a bound argued rather than enforced.

Every subsequent result is a corollary of Theorem 8.2 applied to a well-chosen S . That is the whole of the analysis: a bound that must be re-argued for each new attack is a bound that will eventually be re-argued wrongly.

8.3 Identities are free and worthless

Corollary 8.4 (Sybil invariance) *Let x be an account with no incident stakes. Then $\text{Cap}(S \cup \{x\}) = \text{Cap}(S)$. In particular, an account created by an unheralded key has capacity zero, and creating N of them adds nothing to any coalition.*

Proof. $\partial(S \cup \{x\}) = \partial S$, since x contributes no stakes to the boundary, and $x \notin \mathcal{U}$ since a supply is seated only by a ceremony (§7.4) and no ceremony has seated x . The two sets induce the same cuts, hence the same maximum flow. $\square$

There is accordingly no admission decision to make and no rate to limit, which is also why the ledger carries no creation transition to rate-limit: a row is seated by the first trade naming its key, and what bounds it is the bond that trade already pays (§12). Account creation is permissionless because it is inert.

8.4 Fabricated trade is worthless

Corollary 8.5 (Wash invariance) *Any sequence of acceptances and settlements between members of S leaves $\text{Cap}(S)$ unchanged, regardless of the amounts involved.*

Proof. Such settlements modify $w(c, d)$ only for $c, d \in S$ by Definition 4.2. Those stakes have both endpoints in S , so they cross no cut into S and ∂S is untouched. By Theorem 8.2 the bound is unchanged; and since the flow must still cross ∂S , so is the maximum. $\square$

Corollary 8.5 is why this design contains no wash-trading detector, no volume-anomaly heuristic and no rate model: fabricated settlement is not detected and not penalised, it is arithmetically void. Remark 4.3 explains why a design measuring volume instead of stake cannot have this property: settlement demands two signatures and no delivery, so its volume is free, and a free quantity placed in ∂S makes the right-hand side of Theorem 8.2 unbounded.

8.5 Collusion does not multiply

Corollary 8.6 (Collusion bound) *Suppose every stake from S to its complement has its other endpoint at a single account m , and m is not an underwriter. Then $\text{Cap}(S) \leq \text{Cap}(\{m\})$.*

Proof. Every path into S passes through m , so $\{m\}$ is a vertex separator. A feasible flow into S is therefore a feasible flow into m , and the maximum of the former is at most the maximum of the latter. $\square$

A member who backs an adversary passes on exactly what they hold and no more, however many accounts they distribute it across or however large the stakes they write. The community's exposure to a coalition is its exposure to the honest members who chose to back it, which a community can reason about without knowing who is colluding.

Corollary 8.7 (A coalition cannot underwrite itself) *Appointing members of S as underwriters does not raise $\text{Cap}(S)$.*

Proof. Definition 4.5 draws supply only from $\mathcal{U} \setminus S$, so an underwriter inside S contributes no supply arc to S 's own measurement. Their stakes on other members of S are internal and excluded by Corollary 8.5. $\square$

Remark 8.8 (What this corollary does *not* say) *It is a statement about the members of S , and the credit a coalition wants is issued to the accounts S backs, which are outside it and to which S 's supply arcs therefore do count. That is not a gap in the corollary but a question it does not answer, and the answer comes from elsewhere: a supply is seated only by a ceremony (§7.4), so S cannot appoint its own accomplices in the first place. A rule that let a member declare against their own capacity would leave this corollary true and the community holding 2,048 times its seed in ledger-labelled insured credit (§4).*

This is what lets the underwriter role be open to every member without opening a door: a coalition may propose as many of its own accounts as it likes, measured as a coalition it gains nothing even if the community endorses them, and measured individually its members are still bounded by what flows in from outside.

Corollary 8.9 (Bounded by declared underwriting) *For any set S , $\text{Cap}(S) \leq \sum_{u \in \mathcal{U} \setminus S} \text{sup}(u)$. In particular an underwriter's total exposure — across every member they back, however many — is at most their own declared supply.*

Proof. The supply arcs of $\mathcal{U} \setminus S$ form a cut. $\square$

Corollary 8.9 is what makes the recursion in Definition 4.2 terminate, and it is the community’s aggregate solvency statement: total insured credit never exceeds what its underwriters have collectively declared. That total is not a genesis constant — it rises when the community endorses an underwriter and falls when one withdraws, subject to the floor of §4 — and it is a measure rather than an upper bound precisely because a supply cannot be declared against standing the community itself conferred: under a rule that allowed one, §4 measures the divergence at 204,900 declared against an aggregate of 100. What the community can pay is what its underwriters can pay from outside the set being measured.

8.6 The cost of attack

Corollary 8.10 (Linearity) *An adversary who acquires genuine backing worth B — by trading honestly, at honest prices, with members willing to stake on them — obtains at most B of capacity across every account they control.*

Proof. Take S to be the adversary’s accounts. Their genuine backing is exactly ∂S ; apply Theorem 8.2. $\square$

The proof rests on one word, *genuine*, and that word binds the transition function: it holds only because **no transition writes an edge of ∂S without the signature of somebody outside S** . By Remark 3.3 the sole writer is settlement, which takes the creditor’s signature, and a creditor outside S signing is the adversary having persuaded an outsider to back them, which is what the corollary counts. That condition is easy to break by accident: a debtor swap looks like a discharge, and a rule that staked it would let a ring of k members passing one accepted claim collect an edge each from a creditor who signed once — $\sum w$ linear in k from one outsider’s signature. The swap therefore stakes nothing (§3), and any transition that touches the stake graph has to answer the same question.

Corollary 8.11 (Seats are linear in backing) *Let S' be any set of accounts containing no underwriter, and let $\text{seats}(S')$ count the rows seated by a sponsor in S' . Then*

$$0.02 \cdot V_{\text{base}} \cdot \text{seats}(S') \leq \sum_{e \in \partial S'} \text{peak}(e),$$

where $\text{peak}(e)$ is the all-time maximum weight of arc e and a supply arc counts at its all-time maximum declaration. Minimising over supersets gives the bound at the cut.

Proof. Every seat holds a flow from the source to its sponsor, of one bond unit, on a reservation map the credit layer never reads (Property 12.2). Consider the net seat flow across $\partial S'$: for a sponsor inside S' the whole amount crosses inward, for one outside it nets to zero, so the net inward seat flow is exactly $0.02 \cdot V_{\text{base}} \cdot \text{seats}(S')$. That flow is at most $\sum_{e \in \partial S'} s(e)$, where s is the seat reservation on an arc, and each such value is at most $\text{peak}(e)$: a reservation is refused where it would exceed the arc’s weight at the time it is taken, and nothing but a reservation ever raises one. $\square$

Counting by sponsor rather than by row is what the statement needs and the honest reading: a row an outsider chose to bring in cost that outsider’s reach and is not the set’s doing, and a row the set seated is, wherever its account then sits. The bound is over all-time peaks rather than present weights, and that is exact rather than loose for a permanent stock backed when it is seated: decay lowers an arc without releasing what a seat took on it, so renewal to an old peak opens nothing. Time appears nowhere in the statement, which is the point — a bound that a wait relaxes is a bound on a rate, and a row is a stock. Charging the supply arc is what

makes the aggregate hold: ten operators each vouched at 500 by one underwriter who declared 2,500 hold 125 seats between them and not 250, because the ten stake arcs sum to 5,000 and the arc they all draw through does not.

There is therefore no leverage anywhere in the system: no configuration of identities, volumes, timing or collusion converts B of real economic commitment into more than B of underwritten credit. The protocol has no attack surface distinct from its participation surface. An adversary who wants more must first persuade more members to back them, which is indistinguishable from becoming a member in good standing, and costs the same.

8.7 The scope, and how it is honoured

Everything above is scoped to one ledger: a cut drawn over one order's accounts never measures a coalition that stood up a second. The scope is narrower than it first appears, because a ledger is not a community — two co-operatives sharing an order are inside one cut, and every result above applies to a coalition spanning both — and it is honoured by communities that expect to trade sharing a ledger. What lies outside the order lies outside the arithmetic; the alternatives were measured (§2.2) and each voids the corollaries above at a stroke.

8.8 What is assumed

Theorem 8.2 and its corollaries assume no honesty on the part of any account, no bound on identity creation, no external registry, and no observation of behaviour. They assume:

1. **Underwriters can bear what they declared.** A supply is a promise; the protocol enforces its consequences (§6) but cannot verify in advance that an underwriter is good for it. A community that lets members declare more than they can bear has mispriced itself, and no later mechanism corrects that.
2. **The order is agreed.** Reservation is a conserved budget, so replicas that disagree on the sequence of acceptances disagree on who holds it (Remark 4.9).
3. **Capacity is computed identically everywhere.** The capacity path is integer throughout for this reason; see §12.
4. **No quantity from another order enters.** A single foreign input — a recognised supply, a discharge authorised by a foreign root — voids Corollaries 8.4, 8.5, 8.7 and 8.10 together. The assumption is discharged by there being no transition that can carry such an input.

Nothing else is assumed, and in particular nothing about the proportion of honest members, the cost of an identity, or the quality of any membrane.

8.9 What this does not defend

A creditor's own judgement. Uninsured credit is unbounded by construction (§4). A member who lends beyond a debtor's capacity can lose everything they lent, and the protocol neither prevents it nor compensates it: the decision sits with the party holding the information.

Backing withdrawn by decay. A member's capacity can fall without any act of their own — because the stakes behind it decay, because those who staked lost their own capacity, or because an underwriter reduced their supply. Standing is a claim on others' present willingness, not a possession, and a community whose members all stop renewing watches its credit fall to zero (Remark 4.10).

Allocation under contention. When several creditors would lend to the same debtor, capacity is taken first-come-first-served: the total is bounded correctly, but which creditor obtains it is decided by transaction order rather than by any notion of fairness, and order within a block is chosen by its proposer. A validator that sees a pending acceptance which would exhaust an underwriter can place its own ahead of it in the same block. It is the one place a validator's private interest touches who is insured; what it can take is bounded by the same cut as everybody else's, and a validator is an institution the community chose. Ordering a block's acceptances by their digest would not remove the lever but auction it: a submitter controls the envelope, so anyone can grind a digest to the front of a block in milliseconds, where today only the proposer reorders; and ordering by a value no submitter knows in advance hands the choice back to the proposer, who assembles the batch. The statement above is the honest one.

Acceptance. No result in this section implies that anybody will accept an obligation in this medium. What is proved is that it cannot be inflated by an adversary, which is necessary and not sufficient; the motive to accept comes from the clearing itself (§3.3) and is bounded by the condition under which clearing is possible at all (§10), and neither is a theorem about an adversary. Nor does any result supply a newcomer with their first willing counterparty (Remark 4.11).

9 Economic position

9.1 No rent on the medium of exchange

Property 9.1 (Absence of monetary rent) *No party earns income from the operation of the medium. There is no fee token, no gas, no seigniorage, no interest on capacity, no treasury and no protocol revenue. Validators are community institutions bearing a cost, not intermediaries extracting one.*

A structural commitment rather than a policy setting, and it constrains the rest of the design: spam cannot be priced by fees, so it is priced by refundable bonds (§6); capacity cannot be sold, so it must be earned; loss cannot be absorbed by a treasury that does not exist, so mutualization rests on a standing, signed liability somebody accepted in advance — a supply declaration, the only thing in the model that can carry a loss.

9.2 Bounded standing, and nothing to hoard

Property 9.2 (No hoardable store of value) *The protocol has no balance. What a member accumulates is capacity, which is non-transferable, non-purchasable, bounded by a cut, decaying unless renewed, and held in other members' stakes rather than in the member's own account.*

A member cannot accumulate the medium, corner it, lend it at interest or exit into it. A large claim is a large expectation of future delivery, and it decays into nothing if the community around it stops trading: what one accumulates is a relationship, and relationships lapse. Capacity concentrates only to the extent that many members individually choose to stake on the same account, and even then Corollary 8.9 bounds the result; no mechanism lets holding capacity generate more.

9.3 Loss, mutualized but bounded

Where a loss falls is decided by the tier, not by an ordering between remedies. An **insured** loss falls on the underwriters whose supply carried it, split by exactly what each arc carried

and bounded by Corollary 8.9; if one of them defaults in turn, on the creditor, visibly. An **uninsured** loss falls on the creditor who chose it, alone (Remark 6.5). There is nothing after that, because there is nothing else to draw on: no lender of last resort. A community that could socialize unbounded loss would have created an unfunded liability on its members, the failure mode this design most wants to avoid, since it converts a credit system into a claim on people who never agreed to underwrite it.

9.4 Autonomy and regulatory scope

Obligations here are commercial trade credit between counterparties who have transacted, denominated in a community unit, non-transferable to third parties as an instrument, and not redeemable for currency. The medium is not a deposit, not a security and not a payment token; it is the bookkeeping of mutual trade credit, made explicit. That describes the mechanism, is not legal advice, and varies by jurisdiction. What the design provides for is auditability: a community can publish its state root, prove any member's position, and export a signed history.

9.5 What conventional money still does better

Money is accepted by strangers; capacity here is worth nothing to a community one has not traded with. Money stores value across decades; capacity decays within epochs. Money is anonymous at the point of sale; obligations here are between identified counterparties within a community. Money can be saved by an individual acting alone; provision here runs through named counterparties, and at long horizons through institutions (§11). This design does not replace money. The claim is narrower: for the subset of economic life that consists of regular mutual trade among producers who know one another, a clearing medium fits better than a scarce asset, and that subset is larger than it is usually taken to be.

10 The replacement thesis and adoption

10.1 The conditional thesis

The claim is conditional, and the unconditional version is false. *If* a set of producers trade with one another regularly, *and* a sufficient share of their inputs is available from within that set, *then* clearing their obligations against one another dominates settling each in national currency: the currency each transaction consumed was only ever a way of moving value the group already collectively possessed. Where the antecedent fails, so does the conclusion — a community whose members buy mostly from outside it cannot clear internally, and no protocol changes that. This is a mechanism for one identifiable structure, dense recurring mutual trade, and not a general monetary proposal.

10.2 The adoption path

The declared path is continuous trade-credit clearing, entered gradually. Members who already extend informal trade credit — pay next month, settle at harvest, run a tab — record it here instead of in a notebook. The commercial relationship is unchanged; the obligation becomes checkable, transferable on the debtor side, insured by the community's underwriters where their standing reaches (§4), and eligible for the consented arbitration of §6. A member's first obligation is one they would have extended anyway, on the terms they would have offered, to somebody they already trade with: the protocol's contribution at that moment is bookkeeping, and its guarantees accrue afterwards. Capacity grows as a residue of that ordinary activity — not applied for, granted or scored, but what the stakes left behind.

10.3 Sizing the founding seed

A founding community has one number it cannot easily revise upward afterwards: the external seed. A supply rises only by ceremony (§4), so the door for a raise is a second ceremony rather than an edit, and §7.4 bounds how fast it may open. The common error is to size it against annual volume.

The seed is not consumed. It is reserved, released and reserved again, so what it insures over a year is the seed multiplied by the turnover rate, which settlement terms set and the seed does not. Measured with one underwriter declaring 10,000 and one debtor: the debtor’s first trade is uninsured, since nobody has staked on them yet, and its settlement writes the stake; from then on the debtor books the largest amount their residual capacity insures at term T , every epoch, and settles each obligation on its maturity epoch. The measured year begins one full term after the first such booking, so no term is credited with the warm-up.

settlement term	insured trade per year	× the seed
30 days	120,000	12.0×
60 days	60,000	6.0×
90 days	40,000	4.0×
180 days	20,000	2.0×
365 days	10,000	1.0×

The shortest row is the floor: a maturity below 30 epochs is refused at acceptance and no governed parameter reaches that field (§3), so 365/30 is the ceiling on turnover for any community. Decay costs the table nothing, by design rather than by the process chosen: an edge never decays below its live reservation (§4), so the stake behind an outstanding obligation cannot fade under it, and the settlement that releases the reservation re-stakes the edge in the same epoch.

The quantity to found on is therefore **peak simultaneous insured credit, never annual volume**: a community expecting 100,000 of trade a year on monthly terms needs a seed near 10,000, not near 100,000.

The seed also bounds membership. A row is seated by one bond unit of flow held on the seed’s reach and never released (§12), so a community seats $\sum$ sup over the unit rows and then nobody until a ceremony: at the genesis unit, one row per 20.00 of seed, so a founding membership of M needs a seed of at least M units before any credit is sized. The rehearsal (`the_seed_sizing_rehearsal`) seats founding members by trade, arrives newcomers through members with standing, retires members at the founder’s churn, runs the seed ceremony on the cadence the founder names, and reports the seats beside the credit: at its default process a seed of 5,000 seats its 250 rows and refuses the next thirty-nine epochs into the measured year, while 20,000 seats every arrival. The rate headroom of §7.4 does not carry over between epochs, so a ceremony a month grows the ceiling by 27% a year and only a ceremony every epoch compounds at β .

Who seats a row is a fact about standing, and standing is earned by owing. A seat is held on its *sponsor’s* reach and selling earns none, so the member best placed to bring a newcomer in — the shopkeeper letting them run a tab — can seat nobody until they have borrowed and repaid themselves, and beyond the one self-act their seat paid for — registering guardians, once (§12) — administer their own account only with a co-signer who has. Onboarding and a member’s own administration concentrate in borrowers and underwriters, and a founder should expect to seat the first members themselves. A community on a seasonal rhythm sizes the decay ratio before the seed: at the genesis ratio an unrenewed edge keeps 12% after a quarter and 0.02% after a year (§4), and the rehearsal prints the figure beside the seats.

The two errors are not symmetric. Understating the seed costs friction: more obligations fall to the uninsured tier, which works and is how the network bootstraps. Overstating it has

no later correction, because the one assumption the protocol makes is that underwriters can bear what they declared (§8). Err low, and re-estimate against what the ledger measures: utilisation ($\sum$ reserved over $\sum$ supply); the count of obligations that fell to uninsured for want of capacity — against a ceiling of 15,000 with members each wanting 2,500, six members refuse none, ten refuse four, twenty refuse fourteen and forty refuse thirty-four; and the realised loss rate, the actuarially correct input and the only one that says whether the declared supply was priced right.

Unlike the validator set, the underwriter set is *not* frozen at a founding: a community whose credit could only originate with its founders would wind down as they aged out, since a stake decays and a supply does not renew itself (§4). Members enter the role by an endorsed amendment and leave it by withdrawing, floored at the flow already committed through them. Opening the role keeps a community alive without making it richer: for that, an underwriter has to bring something in from outside, which is a second ceremony (§7), or the community has to merge with another.

Who the ceremony is for. An insured loss is paid in the underwriters' own obligations (§6.1), so insurance is worth what those obligations buy, and that is what a seed ceremony decides. A community sizes its seed in underwriters whose obligations circulate — a wholesaler its members buy from, a cooperative that sells to them, a municipality that takes them in fees — and an underwriter nobody buys from insures little however much it declares. The first assumption of §8 is the same sentence from the other side.

Why anybody underwrites. The ledger pays no rent (Property 9.1) and does not compensate the role; four things outside the medium's operation do. The vote: underwriters are the electorate, and nobody else is (§7). The trade: an underwriter is an institution whose members' or customers' credit it wants to exist — a wholesaler behind its retailers, a cooperative behind its members — which is why suppliers extend terms at their own risk today. The price: a premium is an ordinary obligation from the insured to the underwriter, which the ledger cannot tell from a trade and treats as one in every respect, standing included; settled, it stakes the underwriter on the debtor, bounded by the underwriter's supply like any creditor's stake, so it is no new lever. The precedent: mutual guarantee societies have filled exactly this role for decades, on membership fees and public backing.

10.4 Acceptance, and what actually binds

The question usually put here is whether a producer will accept an obligation in this medium for real goods, and the usual answer — that nothing in the mechanism compels it — describes a different mechanism. A sale discharges before it creates (§3.3), so a producer carrying obligations is paid by ceasing to owe, and the motive is the one that made them a debtor. Measured, netting a debt of 80 against one's creditor writes the whole 80 of stake, where clearing the same 80 through the cascade writes none (§13).

The argument does not reach a member who owes nothing and buys nothing from the community. For them a claim is what §9 says it is, an expectation of future delivery, worth taking only if there are things worth buying from members at prices worth paying. That member is the one the thesis's antecedent already excludes: **the binding constraint is the antecedent, not the willingness**. What share of a community's inputs can be bought from within it is a structural property of that community, ascertainable before a line of this is deployed, and it decides both whether clearing dominates and whether anybody has a motive to accept. A community that satisfies it does not have to be persuaded; one that does not cannot be, by this or any mechanism, and should not adopt one.

A smaller cost sits beside it: the newcomer’s first trade (Remark 4.11). Somebody must extend uninsured credit before capacity exists — a shopkeeper letting a new neighbour run a tab — and that is what writes the first stake. The design bounds that risk and states it; it does not remove it.

11 Intertemporal provision

A medium that cannot be held must still let a member provide for later. Two instruments answer, both already on the ledger: the claim, and capacity. What is not offered is a third, an undated, counterparty-free stock — a *balance*. A balance is a claim with the debtor’s name erased, and every mechanism here consumes the name: insurance is a reserved path from named underwriters, recourse is substitution onto them, discharge is a named debtor’s delivery. Erasing the name removes not the risk a claim carries but the parties who priced it (Property 9.2).

11.1 The claim: provision to a date

A claim is dated, and the date is a term of the trade: a maturity between 30 and 10 000 epochs — decades, at the genesis epoch length — movable only forward, by extension on both signatures (§3). Its amount does not decay: decay is a property of stakes, and the reservation behind an insured claim floors the edges it runs over (§4). Insured, within the horizon below, it survives its debtor: substitution makes the holder whole in claims on the underwriters, one layer and no further (§6). Nothing bounds what a member may be owed — the capacity gate sits on the debtor, because debt spends the community’s underwriting and being owed spends nothing — so a creditor position of any size is expressible. The ledger keeps the claims themselves, never a signed net position.

Provision to a date is therefore available, and priced. For the life of the claim the debtor’s capacity stays reserved and the underwriters’ supply stays drawn (Rem. 6.2); the holder pays nothing while the two parties not holding pay continuously, so a long maturity is a real concession, and its price has a place to live, the face amount agreed at acceptance. The underwriters are asked once, collectively and in advance: a claim is insured only within the *insured horizon* of its acceptance — 365 epochs at genesis, a governed constant between the maturity floor and the maximum horizon (§7) — and a claim booked or extended past it is the creditor’s own, on the creditor’s own signature. A claim meant to run for decades is held as a series, settled and re-accepted at each horizon, each acceptance priced against the community’s backing at that time, which is where the underwriters are asked again. At community scale the same fact is a budget: every long-dated insured claim occupies flow current trade would otherwise be insured by, so provision and trade compete for the same cut, visibly rather than through an aggregate that silently inflates. A saver draws down the way anything is paid here: by buying from their debtors, which nets first (§3.3); by their claims following trade, as debtors’ own sales route obligations onward (§3); by delivery, which settles.

11.2 Capacity: provision across seasons

Capacity is the other instrument, and the partial one: a member who borrows and honours accumulates standing that lets them draw on the community later with nothing in hand. It decays unless renewed, so it provides across seasons rather than decades; it is not transferable, so it cannot be given to a child or left in a will; and it is a claim on a community’s willingness, not on a stock of anything.

11.3 The net saver’s ledger

A member who produces more than they consume accumulates claims — others’ obligations to them — and only claims: selling confers no standing (Rem. 4.4), so their capacity grows only on the side where they buy and honour. That makes visible what a currency balance conceals: one person’s savings are other people’s obligations, worth what those people deliver. The risk is allocated as everywhere else — an insured claim is covered up to what each underwriter declared and no further, an uninsured one is the creditor’s own (Rem. 6.5).

11.4 The horizon, and institutions

No claim outlives the graph that prices it. At a horizon of decades the residual risk is not the mechanism but the world: the debtor, the one layer of underwriters behind them and the community itself must still be there, trading, at the date. That risk is irreducible in any closed credit system, and a balance would carry it identically, with the names, the insurance and the price removed. Provision across decades — pensions, illness — therefore belongs to institutions: a pension society or mutual fund is a member like any other, holding an account, declaring a supply and so standing behind credit with a liability the ledger enforces, nameable as arbiter or beneficiary, measured by the same cut. The protocol supplies that substrate; creating such institutions belongs to the community, which should plan to serve this horizon with them and with assets held outside the medium.

12 Implementation

12.1 One node binary, and a wallet

A validator runs one program — a Byzantine-fault-tolerant consensus engine, the replicated state machine, a write-ahead log and an HTTP surface for reads and submission, in a single binary with no sequencer, relay or indexer — small enough for a community institution to run without specialists. Validators are the community’s charter institutions rather than a market: the validator set drives both quorum and proposer selection, so a runtime change to it must reach the wire and not only the ledger.

A member’s device runs a *wallet*. Keys and signing stay local; the client reads and submits over the HTTP surface of a node its member chose from a declared *set* of node URLs rather than one, since a cross-check whose peer list comes from the party being checked is circular. The node publishes the state commitment, so a client can put the same question to several nodes and compare: a node can withhold or omit, and cannot forge. A handset is not asked to validate — it sleeps, roams and is reclaimed at the operating system’s pleasure, none of which a quorum should depend on.

The client’s own acceptance rule (§4) is signing, so it decides for exactly as long as the client is *running*: in front of the member, or behind, where the client holds its page up against the platform’s own attempt to pause it — a tray the window hides into, and on Android a foreground service under a re-resumed view. For no longer than that. Nothing outside that page holds the key, so a closed app signs nothing, and neither the service nor the tray can decide anything on its own. A device left unplugged and stationary with its screen off has its network suspended by the system, which the client documents rather than fights: it acquires no wake lock, and the exemption is the member’s to grant in their own settings.

Consensus is *more* necessary under this model than under a purely subjective one: capacity is a conserved budget over a shared graph, and the total order decides who lends (Remark 4.9).

12.2 Feeless writes

Writes are free at the point of use (Property 9.1). A member has a per-epoch allowance of 32 transitions; beyond it a transition encumbers a refundable bond of $0.02 \cdot V_{\text{base}}$ against their own write headroom, released after 1 epoch (§6). The allowance is not a per-key entitlement: it goes to accounts the community has put something behind, because keys are free and N keys would otherwise carry N allowances — Assumption 8.1 one layer below credit. A seated row does hold one self-act its seat paid for — registering guardians, once, on its own key — spent by the gate whether or not the write is accepted, and by a refusal alike; seats are bounded by the seed, so it is a stock and never a rate.

Property 12.1 (The write floor is measured against the seed’s reach) *Write headroom is the maximum flow of §4 computed with each underwriter’s source arc carrying only the part a ceremony seated, plus the member’s own such part, gross of outstanding credit, net of what they owe and have already encumbered. It is not the declared supply.*

Gross of outstanding credit, and only the member’s own debt nets. A credit limit is netted because a unit of standing backs one obligation at a time (Def. 4.7); a write floor is what a member has *to lose*, and the only netting it needs is the subtraction the property names. Reading the reach on the residual graph subtracts the member’s own debt twice — their obligation reserves the arcs into them, and what they owe comes off again — and everybody else’s once: a member reached for 500 who borrowed 200 reads 100 of headroom where the property says 300, and at 20% community utilisation a member who has borrowed nothing loses their whole floor and allowance because another member drew the underwriter that reaches them; drawn to its ceiling, such a community stops trading altogether, where a ceiling that binds withholds *insurance* (§4) and the trade it refuses to insure is the trade that dissolves the bootstrap. The qualification the allowance turns on is the same reading taken gross — what the community has put behind the account, not how much of it is unspent this afternoon — with an open default holding the allowance until it is cured, since a default consuming the defaulter’s standing is the sanction.

The distinction from the declared supply is load-bearing. A declaration is a promise made on the strength of another promise (§4), so a chain of them inflates geometrically: measured on six founders at 2,500 with a single honest 300 trade behind the entire chain, write headroom read from declarations **doubles with every accomplice** — 19,200 at seven, 614,400 at twelve — while the real cut over the whole coalition is 300 throughout; accounts being free, that is an unbounded channel rather than an amplification, and it makes the traffic bound vacuous. Read from the seed’s reach, every member of that coalition holds exactly 300, so k colluding accounts hold what k honestly-backed members hold: attacking the write surface costs what participating costs, the same standing this design claims everywhere (§8).

A supply *reduction* is free, because a member must never be priced out of reducing what they stand behind; nobody may raise one (§4), and restating an unchanged supply is bonded like any transition that shrinks nothing. Every free class must answer “how many can one member force in an epoch?”, since a free transition is unlimited and each admitted transaction is recorded against replay: a partial discharge or cure is at least a 128th of the original, so a row is discharged in at most that many payments and the closing one — an uninsured amount is bounded by nothing but the ingress ceiling, so without the floor one allowance slot buys 2^{51} free payments of one minor unit, each a durable replay id and, on an insured row, a re-hold on every validator; transfer and exit close what they touch; the permissionless cranks are idempotent and forget the id of a call that found nothing to do; an assent or an arbiter’s attestation is one per pair, the second refused.

Refusal is a write too. A free class refused at dispatch is priced by one allowance slot of the first signer in canonical order who has one, and where nobody present has one it leaves

nothing behind: the signed intents whose success is a matter of timing — an exit while owing, a lowering below the floor — keep their id where somebody could pay for it, so the identical bytes replayed once they would succeed are refused, while a refusal nobody can pay for is not a durable write. The three permissionless cranks carry no signature to defer and always leave nothing; they are one list, and the node’s ingress asks the same list to decide what may arrive unsigned.

What justifies a zero is whatever refuses the *second* call, not what the first accomplishes: an EXIT that returns success on an account that has already left is admitted 500 of 500 times with no allowance and no headroom. The replay record is why it matters. It is durable, committed state kept for a whole transaction lifetime, so *recording one is itself a write*, and a write no bond prices is the attack the schedule exists to refuse. The gate’s order is therefore load-bearing — charge, then record, then dispatch: a transaction refused at the gate was never admitted and leaves nothing behind; one refused at dispatch has been attempted against the ledger’s economic state, and spends its identifier so that a signature cannot be held until the same attempt would succeed.

Account creation escapes the question by not being a transition. It cannot be bonded on its own — a key that is not yet a member has no headroom to encumber, and a creation somebody had to pay for would be a creation somebody could refuse — and an unbonded transition is bounded by nothing, so the most it could carry is a ledger-wide per-epoch cap, which is a censorship lever rather than a quota: whoever fills it first shuts onboarding for everyone until the boundary, at no cost. An account row is therefore seated by the first *bonded trade* that names its key (§4). Creation stays unapprovable, because the trade decides whether the row appears and nobody else does, and it remains inert (Corollary 8.4), which is why nothing needs approving.

A write budget bounds a rate; a row is a stock. That the trade was bonded bounds how fast rows appear and not how many: a bond is reserved and released, so what it buys over time is unbounded, while a row is permanent and hashed into every state root. Measured, a row that can seat on its own headroom carries a farm behind a single honest edge of 500 from 91 rows after six epochs at a wash of 40 to 1,120 at 100, and raising the establishment floor tenfold leaves every count identical: the floor withholds the *allowance* a seated row would hold, never the row.

Property 12.2 (A row is a stock and holds its price) *Seating a row reserves one bond unit of flow from the community’s seed to the sponsor, over the same stake graph credit runs on, on a second reservation map credit never reads — the sponsor’s own supply arc included — and no transition releases it.*

Both halves are load-bearing. Held per account, the price is the same backing read once per child, so a farm behind one edge compounds; held on the shared graph, a child’s residual is the cut less what its parent and siblings already drew. Released on any schedule, the price is a rate again — a seat shrinking with decay is renewed at 2.3% of the backing per epoch by an accomplice’s free trades — and never released, the count is bounded by backing alone (Cor. 8.11). What it costs: 25 rows behind one edge of 500, 250 behind 5,000, unchanged over 400 epochs of daily renewal, and +25 for a second backer of 500. A community seats $\sum \sup$ over the bond unit and then nobody — the honest preset’s seed of 7,500 carries 375 rows — until a ceremony raises the seed, which is the door every supply arc already comes through.

The sponsor’s own supply arc is in the query, the one place the write layer must not read credit’s rule: a cut into an account excludes that account’s own declaration, because nobody underwrites their own borrowing, so a founding underwriter’s capacity is zero, and a community whose first members nobody could seat would not start.

Three things bend, and are stated rather than worked around. The seat is the one bond no transition returns, because what it bought does not go away by a transition; only the sweep retiring an empty row returns it (§4), which keeps a community with ordinary churn from holding its dead at its ceiling. The ceiling is real: at $\sum \sup$ over the bond unit nobody seats until a ceremony, and a bound that never binds is not one. And the reservation does not reroute earlier seats, so a seat can be refused that a full re-solve would have fitted — the first-come conservatism the insured tier already carries, and never a seat the bound forbids.

Measured against the ordinary acceptance beside it, the seat adds about a fifth: the same trade naming a key that already has a row against one that does not reads 29.7 ms and 35.5 ms at 20,000 accounts, 1.03 ms and 1.18 ms at 1,000, both conditions interleaved in one sitting — less than the two full queries its shape suggests, because both stop at one bond unit rather than computing a maximum.

12.3 Determinism

Capacity is consensus-critical: replicas that compute it differently disagree about whether a transaction is valid. Three commitments make it reproducible.

Every amount is an integer. Stakes, reservations, contract amounts, cached debt, bond encumbrance and arbitration attestations are held in minor units, 100 per unit of denomination, and the maximum-flow computation is exact integer arithmetic. Floating point is *structurally* unsafe here, not merely imprecise: in the flow search, two replicas comparing residual capacities against an epsilon can agree on the resulting value and still take different augmenting paths, leaving different reservations behind; in the book, a cached total maintained by addition and subtraction drifts from the same total recounted, so an audit comparing the two must either demand an exactness the quantities cannot deliver or admit a tolerance, and a tolerance is a claim that the cache and the book may legitimately differ, which they may not. Real numbers cross the boundary in exactly two places: a transaction’s payload on the way in, and a served view on the way out.

The boundary has a ceiling, and it sits where the crossing stops being exact rather than where the integer type gives out. A payload’s amount is a float, so a view leaving and the payment that closes the row against it make the round trip twice — two roundings, hence a discrepancy of at most 2^{-52} of the amount, under half a minor unit only below 2^{51} ; above that a debtor paying the exact figure the ledger served them is told they have over-paid. So an amount over 2 251 799 813 685 247 minor units is refused at every ingress, and refused *before* the conversion, since rounding a real number to an integer clamps at both ends and a clamp is not a refusal.

Iteration order is a function of state. The flow network is built from ordered maps in key order, so adjacency order depends on the ledger’s contents and never on the order records arrived.

The augmenting search is iterative. A recursive walk is bounded only by the level graph’s depth, which is bounded only by the account count, and a stack overflow inside block validation is a liveness fault, so the search maintains its own path stack.

12.4 Cost

There are two costs, and only one of them is a capacity query.

Per acceptance. Booking an obligation runs exactly one query, limit-capped: an acceptance asks whether room exists for one amount rather than for the maximum. It costs 0.17 ms at 1,000 accounts, 2.9 ms at 10,000, 7.1 ms at 20,000 with 160,000 stakes and 36 ms at 100,000 with 800,000; the read side — an account’s published capacity — asks for the maximum, at 0.28, 4.6, 12 and 104 ms. Early termination is worth 1.6 to 2.9 times, an implementation detail rather than a strategy.

Most of an acceptance is the BUILD rather than the search — two thirds at 1,000 accounts and nine tenths at 100,000, since an early exit stops long before the search has seen the network the build had to write; of a full cut the build is a third. The residual network is constructed fresh on every query, so it holds no state that can drift from the ledger. A persistent adjacency beside the edge map is not taken, and not for determinism — held in key order, a patched adjacency and a rebuilt one are one structure a probe can hold equal — but for what it buys: a second structure every write to the edge map must keep equal to the first, in exchange for the build’s share of an early-exit query on a term that bounds nothing, since an acceptance is paid per transaction and the full cuts, which gain a third at most, are the ones the memo below keeps off the block. What is taken is the constant factors that need no second structure: the adjacency is a compressed row rather than a vector per node and the node index is counted rather than sorted, together a factor of three on the same term, measured A/B on one machine; and the reservation on each edge is read by one walk of the reservation map in the edge map’s own order rather than a lookup per edge, without which the build grows with how much of the graph is reserved — from 4.4 to 20 ms at 20,000 accounts between no edge reserved and every edge reserved, and from 33 to 81 ms at 100,000, against 5.6 and 44 ms with every edge reserved once the map is walked.

Per block. Every validator re-audits the invariants of §13 after every committed block, and the cut bound is evaluated over sets: all insured debtors together, and one set per underwriter carrying any of them. Computing that family costs $U + 3 \text{ full}$ queries whatever the block contained, where U is the number of underwriters with live insured debt — 4 ms, 464 ms and 2.13 s for 1,000, 10,000 and 20,000 accounts with $U = n/100$, two orders of magnitude above the acceptance query and, at the top of that range, more than a whole consensus round. That is what the definition costs. A validator does not pay it, for two separate reasons, and the second is the one that matters at an epoch boundary.

The invariant must hold over the whole family after every block; it does not follow that every set must be recomputed after every block. A cut is a maximum flow over free with pristine reservations, so it is a pure function of the stake edges, the declared supplies and the id space, and monotone in all three: raising an edge, raising a supply or admitting more accounts can only raise a maximum flow. A cut this audit has already verified therefore stays a valid *lower bound* until one of those three falls, and a draw within that bound closes the clause with no query at all; only a bound that fails to cover the draw costs a real one. What can lower them is a short structural list rather than a hope about traffic: Definition 4.2 is a peak rule, so settlement — the only writer of the graph — can only raise an edge, and the whole of the other direction is decay at the epoch boundary, a governed re-denomination, and a supply reduction. So a block that changed nothing economic computes **no** cut at all, and a block drawing new insured credit through one underwriter computes **two** — the set of all insured debtors, which gained a member, and that one underwriter’s.

That argument is worth exactly its premise, and the premise fails once a day. **Decay lowers every edge free to fall**, so the epoch-advance block finds the monotone argument gone and pays the whole family: 201 cuts after decay against the 201 a cold audit computes. The memo buys nothing on the one block that was already the odd one out.

The second reason: an obligation is a witness. Invariant 3.6 asks whether a set’s drawn credit is within its cut, and a cut is a *maximum* flow, so a feasible flow of that value settles the clause without producing the maximum. Every insured obligation already stores one: exactly which arcs it drew and how much on each (Remark 4.8). Feasibility is three conditions, and the audit checks two of them anyway — arc usage within the stake (invariant 4) and supply usage within the declaration (invariant 5); the third, that the stored hold is a *flow* at all — what arrives at a node leaves it — is one pass over the arcs the obligations hold, with no query in it. **Decay cannot touch a witness:** invariant 4 floors every edge at its live reservation, so whatever the epoch did, the arcs the holds occupy are still there. The memo dies at the boundary and the witness does not, which is precisely where the cost was.

Measured (**just cost**), both arms on the same state and adjacent in time — the cold definition, which computes every set, against the commit path a validator runs:

accounts	U	definition	commit path	sets, computed
1,000	10	4.2 ms	1.0 ms	11 / 2
5,000	50	148.9 ms	6.0 ms	51 / 2
10,000	100	544.3 ms	13.3 ms	101 / 2
20,000	200	2,674.6 ms	28.4 ms	201 / 2

94× at the top of the range, and the two cuts still computed are invariant 6’s forward-and-reversed determinism pair rather than the cut bound, which computes none. The family has not shrunk — the invariant still holds over all 201 sets after every block — and neither has the theorem; a stored proof settles it instead of a computation.

The audit is not the only whole-ledger pass a block pays for. A block also commits to the ledger, and a full build of that commitment re-encodes and re-hashes every row. The tree is incremental instead (the state root, below): a leaf’s salt binds the EPOCH rather than the block, so an unchanged record keeps its leaf hash for an epoch and an ordinary block rehashes the rows it wrote and the paths above them. Measured as the pair in one sitting on one machine, over the shape a wash farm produces — one account and one in-edge per seat — the definition’s full build against what a block costs through the incremental one: 1.1 ms against 0.007 ms at 1,000 rows, 6.1 against 0.009 at 10,000, 30.6 against 0.012 at 50,000, 73.6 against 0.015 at 100,000. The definition is linear in the ledger and the block is flat in it; what the block scales with is the OUT-DEGREE of the accounts it wrote to, because a stake row is one leaf per creditor, so writing an edge re-encodes that creditor’s own row and nothing else.

One full build remains, at the epoch boundary, where every salt changes — the block that already rewrites every member row, decays every stake and sweeps the contract book, so it costs nothing the boundary was not paying. What the epoch-bound salt buys is a bounded disclosure rather than none: within one epoch an unchanged record keeps its leaf hash, so a member holding two of their own proofs from that epoch learns from the sibling hashes whether the record adjacent to theirs in identifier order changed between the two heights; across epochs nothing links, and a proof is served only to its subject or to a validator, who holds the ledger anyway.

No working copy is kept per block. The invariant audit is a fail-stop whose depth is the log append that happens after the audit: a refused block is not on disk, and the node that refused it stops, commits nothing further and answers every read with the height it stopped at and why. A copy to roll back to is a second whole-ledger pass per block for the window between the refusal and the process exiting — 0.1, 1.2, 17.6 and 19.0 ms across the same four sizes in the same sitting — and what remains linear per block is the audit’s own structural walks.

The witness is the data and never a flag: the audit compares the present edges and supplies against its own snapshot element by element, because a transition that lowered an edge is

exactly what the audit exists to catch. None of it can change a verdict — it decides only whether a dominated computation runs — so a validator resuming from a snapshot starts cold, does more work, and halts on exactly the states a warm one halts on. The order the two are consulted in is load-bearing: asked before the memo, the witness runs on every audit and an ordinary block pays for a pass it did not need (27 ms against 67 ms at 20,000 accounts); asked only where a set is about to cost a real query, it is free on exactly those blocks, and what it proves is banked as the entry’s floor.

Held at 20,000 accounts and varied in U instead, the *definition’s* cost is linear in the underwriter count — 0.35 s at $U = 20$, 2.02 s at 200, 3.14 s at 400 — and that linearity is what the witness removes: 4.2× at $U = 10$ against 94× at $U = 200$ in the table above, because the arm that grows with U is the one the commit path does not run. That matters because the underwriter role is open by design, and an underwriter seat carrying live insured debt is a stock: four priced writes seat it, on a young community inside the free allowance, and only settlement retires it — an expired obligation holds its arc indefinitely, and a declaration is floored at the flow committed through it (Rem. 6.2). Three other shapes are available and none is taken, because each changes what the ledger means rather than what it computes: capping U closes the role, a standing fee for a seat is rent (Property 9.1), and moving the audit off the commit path changes what a commit guarantees.

Measured, **just cost**, on an Intel Core i9-14900HX under `--release` at a one-minute load average of 3.8, which the harness prints beside every figure; the harnesses are `crates/kernel/tests/cost.rs` and `crates/state/tests/cost.rs`, and the community is $n/100$ founding underwriters with eight stakes per account. A wall-clock figure without its generator and its machine is not a measurement, and on a loaded box the same harness reads five times higher, which is why the table above measures both arms on one state adjacent in time: a ratio taken that way survives a box that is not idle, and a difference taken against a figure from another run does not.

Both costs are bounded by *community* size and never by network size, so a federation of any extent costs each community exactly what its own membership costs.

12.5 The state root

Every block commits to a salted Merkle root over the ledger, with inclusion proofs for any record. The ledger is divided into seven sections — accounts, obligations, proposals, validators, one stake row per account, the replay window bucketed by expiry, and one scalar leaf for the rest. A stake row carries, per debtor, the arc’s weight beside both reservations on it — what outstanding credit holds and what the seats do — because the two are facts about one arc and a section of their own would rehash a second tree for the same writes. Leaves are length-framed in every variable-length component, commit their own index, and use distinct leaf, inner and section domain separation; each section commits its own leaf COUNT, once, at its root; odd levels promote the last node rather than duplicating it.

Each of these closes a specific forgery. Unframed concatenation is not injective, so the boundary between key and value can be slid and many distinct records share one preimage. Uncommitted position lets a valid path be replayed at another index. An uncommitted count lets a path be replayed against a tree of another size — and it must be committed somewhere other than the leaf, because which side each sibling sits on is a function of the index alone, so a proof for leaf 0 of a three-leaf section folds byte for byte to the root of a four-leaf claim. Shared leaf and inner preimages let an inner node be passed off as a leaf. Duplicate-last promotion makes two distinct trees share a root. Binding the count at the section root rather than inside every leaf is also what makes the tree incremental: a leaf survives its section growing, so seating an account rehashes one leaf and its path instead of its whole section.

Leaves are salted from a per-chain secret fixed at genesis, so sibling hashes travelling inside somebody else’s inclusion proof do not expose preimages — amounts and identifiers occupy

a small enough space to brute-force otherwise. The salt binds the record’s value through its digest and the EPOCH the record is committed in, so a disclosed salt unblinds one epoch’s value of one record and nothing else, no leaf hash survives an epoch boundary, and within an epoch it does, which is the disclosure the cost paragraph states.

The commitment has two implementations held to each other: the definition is a pure function of the ledger that rebuilds every leaf; what a validator runs patches the rows a block wrote. The two are computed side by side after every transition in the state harness and in the population runs of §13, and a disagreement is a failure rather than a slower path — the discipline the memoised audit is under, except that this one decides a consensus value.

12.6 Privacy model

On the ledger a member is a pseudonymous handle. Charter policy may seal precise amounts to the parties and validators, with other members seeing bucketed magnitudes. The capacity computation is *public* in structure: any observer holding the ledger computes the same capacity a validator does, so the quantity is checkable rather than announced. The stake graph is therefore visible in shape even where amounts are sealed, and a community that regards its trading topology as sensitive should treat that as a known cost.

12.7 Workspace

The economic kernel is a small pure library — capacity, cascade arithmetic, constants, an advisory risk score — with no I/O, no clock and no state, so every function in it is a deterministic map of its arguments. The ledger crate holds the state machine and its invariants; the node crate holds consensus, storage and the network surfaces; the client is a separate application that signs locally and reads a node over HTTP. Genesis values live in the kernel and are cited above through generated macros, with a build gate that fails if the two ever disagree.

13 Verification programme

13.1 Layers

Proved. The results of §8 are proved from Definitions 4.2–4.7 and max-flow/min-cut, and they are short: a bound re-argued per attack is a bound that will eventually be re-argued wrongly.

Enforced. Seven invariants are audited over the whole state on the commit path rather than in tests — after every committed block on a validator, and after every transition in the harnesses — so a violation halts the chain rather than propagating; what that costs per block is §12:

1. the **cut bound** (Invariant 3.6), evaluated over *sets* and never only singletons — all insured debtors together, and, per underwriter, everyone drawing through them. Two details are load-bearing: the cut is measured on a pristine residual, and only supply reaching the set from *outside* it counts on the left, as the network construction drops the source arc of any underwriter inside the measured set; both are exercised by probes that fail when the clause they hold is removed, not only by fixtures in which the audit is expected to pass;
2. **reservation conservation** (Invariant 3.5), exactly, in integer minor units;
3. the **stake bound**. What an edge may carry is what its creditor could confer *at the height it was written* — at settlement, not at acceptance — enforced where it is written rather than re-derived here, because a creditor’s own backing legitimately moves afterwards and

an audit flagging an edge above its creditor’s present capacity would halt a healthy chain; the stored graph is held to every edge naming real accounts and carrying something;

4. the **decay floor**: no edge sits below its live reservation;
5. the **supply floor**: no underwriter’s declaration sits below the flow committed through it;
6. **determinism**: capacity recomputed from committed state reproduces the value the block was validated against, bit for bit;
7. **seat conservation**: the seat reservation maps are exactly the sum of what the live seats hold, every seat is a flow from the source to its sponsor, and a sponsor is a member — which the sweep keeps true by never retiring a row that sponsors a live seat. It claims conservation and nothing else: a seat is deliberately *not* floored against decay or against the declaration it draws through (Property 12.2), and a seat’s *size* is compared to nothing in either direction, because both bounds are properties of the write — a seat takes one bond unit at the unit then in force, a re-denomination floors it, and an amendment of the unit leaves it where it is; a clause comparing a standing seat against the live unit halts every node on the first downward amendment of the one dial the design offers for the seat count.

The same audit checks obligation conservation (Invariant 3.4) — each account’s cached debt against a recount of the book, and the two totals against each other, as exact integer equality — and the structural facts no cut expresses: no contract names one member on both sides, a validator is an active member, arbitration attestations come only from the panel both parties pinned, and the key index round-trips in both directions. Non-negativity needs no check: every stored amount is an unsigned integer, so the type carries the floor.

Searched. A population of behaving agents — honest traders, a deadbeat, a wash ring, a sybil farm, a griever, a governance coalition, a panic exitter, an honest-then-evil sleeper, a late defaulter and a hoarder — runs over the same transition function, and every transition it emits is judged by the same audit. The programme is split three ways on purpose. *Does the mechanism hold under adversarial play* is a search, answered by deterministic strategies over the alphabet. *Does it produce a working economy under plausible behaviour* is distributional, answered by seeded populations against a control that is the same seed with every treatment seat replaced by an honest trader and aged the same number of epochs — advancing an epoch decays every stake, so a control that has not advanced the same number is not one. *Would people join, and what would they try* is generative, and the only part a language model touches: it produces narratives, which are hypotheses, which the two halves above judge. No figure a persona produced appears in this paper, because a generator whose behavioural variance is two orders of magnitude below the phenomenon’s would be describing itself.

Each run is one seed and replays bit-identically, so a violated invariant is reported with the seed, the tick, the agent, the intent, the transaction and the command that reproduces it. Twenty-four scenes are pinned as a gate; a strategy that bites is promoted by hand into a named regression test rather than by replaying its seed, so the regression does not depend on the strategy code that found it.

Measured. Each claim has an adversarial test that fails if the property regresses — gates, not illustrations. Every row below names a probe in the tree: a claim’s *measurement* must exist, not only its wording.

13.2 What has been measured

Two independent implementations answer the capacity question — the kernel that ships, and a reference over `scipy`'s maximum flow written by people who had never heard of this design — and the fixtures demand bit-identical agreement rather than closeness. Both readings are pinned: the credit one, which excludes a target's own declaration, and the write one, which includes it. The reference states the seat bound too, and reaches the same count from one edge of 500 that the kernel does. The rows are grouped by the result each gates.

The cut bound and its corollaries.

Claim	Result
Identities are free and worthless (Cor. 8.4)	Sixty fresh keys, measured together as a set: capacity 0
Fabricated trade confers nothing (Cor. 8.5)	2,000 settlement cycles between two accounts nobody backs, each claiming an unbounded amount: capacity 0
A colluder passes on what they hold (Cor. 8.6)	One honest trade of 300 behind the colluder; forty accounts then pump each other for eight rounds at whatever each may confer: region capacity 300, with and without the colluder in the set
A coalition cannot underwrite itself (Cor. 8.7)	Eight rounds of self-appointment: pinned at what backs the coalition from outside
An underwriter's exposure is their supply (Cor. 8.9)	Backing twenty-one accounts: region capacity 2,500, not 52,500. Twenty sybils, fifty rounds, with back-staking: 2,500
Attack costs what participation costs (Cor. 8.10)	1 to 20 genuine relationships spread over five keys: capacity = stake, ratio 1.00 at every k
Conferring above your own inflow is a no-op	A grant of 10^6 from a member holding 1,000: the newcomer gets 1,000
Selling confers nothing on the seller (Rem. 4.4)	2,000 settled sales to twenty backed members: capacity 0; one honoured purchase confers standing at once
Usage cannot seed insured credit (Ass. 8.1)	1,000 settled trades with no underwriters: no stake written, cut 0, and no supply is declarable
A pool funded from inside re-labels the seed (§6.4)	10,000 levied settlements: founderless pool 0; seeded, members $\cup$ pool = the seed exactly
One ledger hosts many communities (§2)	Two co-ops, one chain, one underwriter each: 2,500 apiece and 5,000 together; one trade across joins them, with no transition involved

Reservation, decay and the reserve path.

Claim	Result
No double-draw across creditors	Fifty creditors lending to one debtor in turn: total equals the cut, not a multiple of it

continued overleaf

Claim	Result
One supply is lent once (Rem. 4.8)	One underwriter of 2,500 backing k debtors, $k = 2, 5, 20, 100$: each looks fully backed and together they lend 2,500
Settlement returns the whole path	A multi-hop reservation released in full, both hops and the supply arc; and 4,000 steps of mixed traffic drain to zero reserved and zero committed
The bound holds under traffic that presses it	4,000 reserve and release steps over a two-hop graph, on the independent implementation: zero violations across 6,054 set checks at both quantifiers, peak outstanding 12,800 against a ceiling of 15,000, a third of the reservations refused
A supply cannot be withdrawn under live credit	Withdrawal to the committed floor holds the bound; one step below it, 15,000 outstanding against a capacity of 12,500
Decay cannot release live collateral	24 epochs at 80% reserved: the bound held every epoch, and a fully reserved stake does not decay at all
Capacity is reproducible (§12)	Integer maximum flow identical across input permutations, and bit-identical to the independent implementation on every fixture
A community starts from zero	First trade of 400 uninsured; it settles; capacity 400; the second trade of 400 is insured and holds all of it

The transitions around the model.

Claim	Result
A routed claim is never downgraded (Prop. 3.1)	400 randomised sale scenes, over 100 of them with an insured claim to lose and more than 10 splitting one across buyer and seller: no creditor's insured total ever fell
A debtor swap stakes nothing (Rem. 3.3)	A ring of k members passing one accepted claim: $\sum w$ from the original creditor is 0, where a rule that stakes the swap measures linear in k — 3,000 across ten members from one acceptance of 300
A drain relieves and does not rebuild (Prop. 3.2)	An obligation of 80 cleared by a supporter writes 0 of stake where the debtor paying the same 80 writes all of it; clearing your own debts through a sale writes 0 where netting the same debt against your creditor writes 80; and three loans of 60 carried leave a member at capacity 200, against 380 honoured

continued overleaf

Claim	Result
A loss lands on the arcs that carried it (§6.1)	The claim splits across underwriters by exactly what each arc held; the debt subrogates to them; the defaulter's standing stays consumed; curing every piece makes each whole and frees exactly what it held
An absorbed loss throttles the declaration (Rem. 6.2)	114 randomised scenes with 204 partial cures, on both acts that reach the re-hold: no throttle ever moved off the underwriter that was substituted
The write floor is the seed's reach (Prop. 12.1)	Read from the declared supply, a coalition's write headroom doubled per accomplice — 614,400 at twelve against a real cut of 300. Read from the seed's reach, k colluding accounts hold exactly what k honestly-backed members hold at $k \in \{3, 7, 12\}$, while a founder and an honest member are unchanged
And it is gross of live credit (Prop. 12.1)	Read on the residual graph the floor falls for everybody else's borrowing and twice for the member's own: reach 500 and a loan of 200 give headroom 100 against the 300 the property states, at 20% utilisation a member who has borrowed nothing loses a full allowance and its whole floor, and a community at its ceiling refuses an ordinary trade, a sale and a newcomer's first trade alike. Gross, the ceiling withholds insurance and nothing else, the $k \in \{3, 7, 12\}$ parity above is unchanged, and a declarable rule runs the coalition's declarations to $171 \times$ the floor
Every write has a payer with standing	2,000 trades attempted by keys with no standing seat nothing and censor nobody, while an honest newcomer's first trade is admitted in the same epoch
A row is a stock priced in the cut (Cor. 8.11)	25 rows behind one edge of 500 and 250 behind 5,000, unchanged over 400 epochs of daily renewal, +25 for a second backer of 500 and +5 for raising the first by 100. The honest preset's seed of 7,500 seats 375 rows and then nobody until a ceremony; a farm behind one honest edge holds 23 rows over 5 ticks and the same 23 over 300; and an honest population with a newcomer stream is refused nothing by the seed — 113 rows over 120 ticks against a ceiling of 375

continued overleaf

Claim	Result
A list-writing transition carries its own bound	A rotation is refused one key over the ceiling and admitted at it. The vector is whatever the transaction carries, so the bound belongs to the transition that writes the list: without it, 100,000 keys are written for a headroom spend of zero
A forfeiture is a reservation that never returns (Prop. 6.6)	Sustained exhaustion forfeits once and the headroom does not come back; an intermittent burst never reaches the sanction; a forfeited member can still discharge what it owes
Governance answers to the ceremony (§7)	Seven accomplices, whose declarations would reach 34,200 against a 15,000 seed and are refused, enact <i>nothing</i> — and neither does a member the founders genuinely backed for 2,500; a founder who holds the whole seed and backs one member for all of it enacts alone, while that member enacts nothing; enacting sets are upward-closed over all 32 subsets of five unequal founders
Who orders the ledger takes two thirds (§7)	Three of six founders — Θ exactly — move a governed constant and move neither the validator set nor the suspension of a validator; the fourth carries both. A member holding no voting power is suspended at the ordinary bar
A validator signs with a key that is not its member key	Voting power is refused for a member with no registered consensus key, and the key cannot be retired while the power stands; the key resolves to nobody as a transaction signer
An amendment answers to the seed (§7.4)	Six joiners whose capacity would carry $64\times$ the seed are refused a declaration, a vote and a unit of amendment headroom alike; an endorsed supply of 100 behind three members is worth 100 each and 100 together; β at its ceiling with every passable amendment passed reaches $4.3\times$ the seed in thirty epochs
A coalition cannot insure the accounts it backs (Rem. 8.8)	Twelve joiners wash-backing each other behind a seed of 100 issue at most the seed in insured credit to the accounts they back, and lose at most the seed when they all default — against 204,800 under a rule that lets a member declare against conferred capacity
A panel decides, not the first quorum to sign (§6)	Five of sixteen arbiters attesting the maximum first do not carry the award: the remaining eleven are still heard, and the median over all sixteen is what mints at the window's close

continued overleaf

Claim	Result
A block costs what somebody paid for	A block over the proposer's own batch is refused before it is applied, and so is one carrying an envelope no required signer signed — which is otherwise refunded, unbilled and re-appliable without limit
The fail-stop survives a restart	A block that breaks an invariant is not written to the log, a store the invariants refuse is not resumed from, and the node that refused it commits nothing further and answers every read with the height it stopped at
The incremental commitment is the definition	The pure full build and the patched tree computed side by side after every transition and every epoch crank, and held equal; with the write record dropped from one edge map the two diverge on the first settlement
Re-denomination preserves every real position (Thm. 7.3)	Every two-decimal amount from 0.02 to 1,000.00 through all four discharge paths under the full audit, at each governed factor; a fully drawn community re-denominates and audits clean
No transition breaks an invariant	300 randomly generated transitions over nine kinds, with all seven invariants audited after every one
The memoised cut never changes a verdict	Cold and cached audits driven side by side after every transition in the tree, and held to the same verdict. With the monotonicity guard removed the cached audit <i>passes</i> a state the cold one rejects — a narrowed funnel, 5.00 drawn against a cut of 4.00, with invariants 2, 4 and 5 clean — so the probe fails exactly when the clause it gates is gone
And a block that changed nothing costs nothing	20,000 accounts, $U = 200$: 0 cuts over sixteen audits of an unchanged state. A draw that outgrows a standing bound still costs its query
Every outstanding hold is a conserved flow	Checked after every transition and every epoch crank in every suite, and not implied by the other six: splitting a partially cured subrogated claim by proportion instead of by route leaves 71,429 exiting an account with an inflow of zero with all six green, because a proportional split of each arc gives a piece arcs its own supply never reached

continued overleaf

Claim	Result
And the audit verifies it instead of recomputing the family	20,000 accounts, $U = 200$, both arms on one state: the definition computes all 201 sets in 2.13s where the commit path computes 2 in 28.4ms. The 2 are invariant 6's determinism pair; the cut bound computes none. A hold that is not a flow is refused the shortcut — the narrowed-funnel fixture states a supply arc it never records, so the cut is computed and the violation still fires
A binding ceiling rations insurance in whole rows	With 2.00 of a supply arc free, one row of 3.00 insures nothing where three rows of 1.00 insure 2.00: first-come, and smaller first if you split
An uninsured default leaves the creditor's standing alone	Measured against the counterfactual rather than before-and-after, since standing decays on its own: two identical communities walked the same 31 epochs, one with a free key defaulting on 250 of uninsured credit, and the creditor's capacity is equal in both
A sponsor stays seated while a row it seated lives (§4)	A member who seats one newcomer and trades nothing for eight hundred epochs is present at every boundary; retired at the boundary after the row it seated, never the same one; a sponsor of five rows stays while the last one lives. Retired earlier, the seat is orphaned and the sweep's own block fails the audit on every node
The seat unit moves without touching a standing seat (invariant 7)	The bond fraction $0.02 \rightarrow 0.01 \rightarrow 0.02$ with a seat standing audits clean at each step; the old seat holds 20.00 beside a new one at 10.00; a re-denomination with seats standing is clean too
A refused free class is priced or leaves nothing (§12)	Sixty-four keyless ROTATEFINALIZE refusals record nothing; a debtor's refused exits spend one allowance slot each, and past the allowance nothing durable is written; a contract, proposal or member id the ledger has not issued is kept out of a mempool
A partial payment is at least a 128th of the original	0.01 and 781.24 against 100,000.00 refused, 127 installments of the floor admitted and a closing payment of any size; without the floor, 3,000 payments of one minor unit for one allowance slot with 9,997,000 to go

continued overleaf

Claim	Result
A partial payment shrinks the hold where it is (§3)	D insured by U_b alone, U_a reaching D since, a payment of 100: the hold reads $\{U_b : 200\}$ where a re-solve read $\{U_a : 200\}$; measured on a re-solve, 0.7 ms per payment at 2,000 accounts and 2.3 at 10,000 against a microsecond
A routed claim keeps the earlier of its dates (Prop. 3.1)	A 30-epoch claim routed through a 10,000-epoch sale matures at 30; a 100-epoch claim through a 30-epoch sale at 30; two claims from one creditor on two dates become two rows on their own dates
Standing is the peak over cumulative repayment (Def. 4.2)	1,000 honoured in two halves stakes 500 after the first and 1,000 after the second, equal to 1,000 paid at once; repeating the obligation raises nothing
The panel survives substitution (§6.1)	An insured claim with a panel defaults; the row becomes the underwriter's, the terms still name buyer, seller and 300; the arbiter attests and the award is minted from seller to buyer at the window's close, never against the underwriter
A suspended underwriter lowers and leaves (§7)	To the committed floor while a claim runs through them, to zero once it settles, then exits; a raise is refused as for everyone
A creditor's guardians discharge (§6)	Two of two cure a 600 uninsured default the silent creditor cannot sign, and the debtor's allowance returns; one is under the threshold; a debtor's own guardians pay nothing
The sanctioned vote on reinstatement (§7)	Half the seed suspends the rest and withdraws; three fifths of what is left, suspended, carries its own reinstatement, two fifths cannot, and neither may propose anything else
The validator floor holds every removal path (§7)	At a floor of four, the exit of a validator, a power of zero and a suspension of a validator are each refused; a fifth validator's exit is admitted
A unit that would round to nothing is refused	A bond fraction or a re-denomination under which the unit is under one minor unit is refused when proposed and, when the base moved since, when enacted

13.3 Negative results

Five designs were measured and rejected; the reasons constrain any future revision.

Local approximation of trust does not localize. On a graph with overlapping community membership, a neighbourhood-truncated computation is accurate only once the neighbourhood has grown to most of the graph: at two hops the capacity error is tens of percent, at three the ball covers 67–100% of the accounts. The ball is a constant *fraction* of the graph, not a constant size, so cost independent of network size is unattainable by truncation; bounding the

community, not the walk, is what works.

Flow cannot originate everywhere, and direction alone does not fix it. With every account an unbounded source, a colluder’s honest 300 compounds to 10,000,300 in three rounds, and directional stakes do not prevent it; supply must be bounded, which Definition 4.1 does.

Unbounded supply makes an underwriter an unbounded trust point. With infinite supply, one underwriter backing 21 accounts confers 52,500 rather than the 2,500 they could confer on any one of them; bounding supply bounds the aggregate.

Settled volume is not a sound measure: it is fabricable by two signatures (Remark 4.3).

Independent evaluation leaks by the number of evaluators: 200× with two hundred creditors, which forces reservation and, with it, a serialization point (Remark 4.9).

13.4 Status

The model is specified and built. The kernel computes capacity as an integer maximum flow and is cross-pinned against an independent implementation; the state layer drives the whole alphabet of twenty transitions and audits all seven invariants after every one of them, on the commit path rather than in tests; the substrate runs — consensus, block production and the state root with its inclusion proofs in one node binary, custody and signing in a wallet client. What has not happened is an external audit, and nothing here claims production readiness: the model has not run a pilot genesis, on-device custody has not been exercised end to end, and no party outside the project has reviewed any of it.

What a green gate claims is read narrowly, because the loose readings fail silently. **A passing suite is a claim about what its tests assert**, and a test can assert a defect — intended behaviour is a description a defect can wear through a release. **A passing invariant is not a claim about what it does not count**: all seven stay green while a discharge destroys a creditor’s recourse, because obligation is conserved by that move and what dies is recourse, which no invariant counts — Property 3.1 has its own probe for exactly that reason. And **a green build is a claim about the gates that exist**: no gate in this tree reads a document.

13.5 Method

Four rules produce most of what the sections above assert; a fifth — that a claim’s measurement must exist and name its probe — is what the tables above are.

Test at the quantifier the theorem uses. The characteristic defect of this class of model is a bound asserted over *sets* and tested over *singletons*: one underwriter’s supply drawn by several debtors, a reservation held along several hops, a supply withdrawn while several obligations stand on it, a write channel measured on one member of a chain. A per-account check passes in every one, and in the last it leaves a whole theorem vacuous — the quantity is a growth rate, and a single point on it looks like a constant.

Measure against the counterfactual, not against before-and-after. Standing decays on its own, so walking thirty-one epochs to reach a default and then comparing the creditor’s capacity with what it was credits the default with the decay. The uninsured-loss probe is therefore two identical communities, the same epochs walked in each, and the only difference the thing under test.

A premise worth measuring may turn out to be structural. The cut memo rests on “most blocks decrease nothing”, which reads as a hope about workloads. It is not empirical: Definition 4.2 is a peak rule and settlement is the graph’s only writer, so no ordinary traffic can

lower an edge, and the complete list of what can be is three transitions. Checking the sentence turns a hope about workloads into a property of the model — the paragraph above, run in the other direction.

When the code and the document disagree, find out which one is coherent. Neither is the specification by default. The arbitral award runs from creditor to debtor because it is the buyer's remedy for non-delivery, which a document can state backwards in perfect confidence; the underwriter who cannot leave is the throttle working, which a document can as confidently call a defect. A comment or a paragraph is the one part of a tree no gate reads, which is where a superseded model survives longest.

14 Conclusion

A credit community must decide how much to stand behind each of its members, and every mechanism for deciding this tends toward the same failure: the decision becomes a process, the process accumulates rules, and the rules accumulate exceptions until nobody can say what the system guarantees.

This design makes the decision an equation. Capacity is the maximum flow from a community's underwriters to an account, over directed stakes that creditors have placed and that nobody can place beyond what they may confer, with outstanding credit reserving the flow that justified it, and an underwriter is any member who has declared a liability they will stand behind. Admission, reputation, credit limits and Sybil resistance are that one quantity, and the security results follow from max-flow/min-cut rather than from a policy anybody has to enforce: identities are free and worthless, fabricated trade is arithmetically void rather than detected, a colluding member passes on exactly what they hold, and no configuration of identities, volumes, timing or collusion converts real commitment into more credit than it is worth, so attacking costs what participating costs (§8).

What it costs: a newcomer needs somebody willing to take a first uninsured risk, and no mechanism supplies that willingness; capacity can fall through no fault of its holder, being a claim on others' present willingness rather than a possession; allocation under contention is first-come-first-served, so the total is bounded correctly while who gets it is decided by ordering; and the medium is not money — not accepted by strangers, not a store of value across decades, not saveable by anybody acting alone.

The boundary this design does not cross is economic, and narrower than the one usually drawn here. It guarantees that a community's medium cannot be inflated by an adversary, that its books balance, that its losses are bounded by what its members agreed to bear, and that its unit can be re-denominated without disturbing a single real position. It does not have to make anybody want to *hold* the medium, because nobody holds it: a sale discharges before it creates, so a member who owes is paid by ceasing to owe. What it cannot supply is the condition that makes this true — enough of what a community buys being available from within it — a fact about the community, knowable before anybody founds anything. Nor can it seed a community that underwrote nothing at genesis: zero is absorbing, and the ceremony is the only door.